

INDUSTRIAL TRAINING PROJECT REPORT
ON
SUMMER INTERNSHIP

Submitted by:

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1/23/SET/BCS/526

Under the Guidance of

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in partial fulfillment for the award of the degree of

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IN
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School of Engineering & Technology

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Declaration

I hereby declare that this project report entitled **PLACEMENT TRACKER SYSTEM** by **SANGEETA KUMARI (1/23/SET/BCS/526)**, being submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in **Computer Science and Engineering** under School of Engineering & Technology of Manav Rachna International Institute of Research and Studies, Faridabad, during the academic year June-July, 2026, is a bonafide record of our original work carried out under guidance and supervision of **Dr. CHARU KATHURIA, ASSOCIATE PROFESSOR** and has not been presented elsewhere.

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**Manav Rachna International Institute of Research and Studies,
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School of Engineering & Technology

Department of Computer Science & Engineering

June-July, 2026

Certificate

This is to certify that this project report entitled **PLACEMENT TRACKER SYSTEM** by **SANGEETA KUMARI (1/23/SET/BCS/526)**, submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in **Computer Science and Engineering** under School of Engineering & Technology of Manav Rachna International Institute of Research and Studies Faridabad, during the academic year June-July 2026, is a bonafide record of work carried out under my guidance and supervision.

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This is to confirm that Sangeeta kumari has successfully completed Training at Astrea IT Services Pvt Ltd from June 1, 2026, to July 24, 2026. The Training was on the Salesforce.com cloud computing development platform. We would like to wish Sangeeta kumari all the best in his/her future endeavors.

Sangeeta kumari has shown great enthusiasm and interest during the training period. Her/His understanding of the concept is good, and he/she has effectively participated in all phases of the training. During this period, his/her conduct has been good.

Sincerely,

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ABSTRACT

The Placement Tracker System is a cloud-based application developed using Salesforce CRM to automate and simplify the campus placement process in educational institutions. Traditional placement management methods rely on manual record-keeping and spreadsheets, making the process time-consuming, error-prone, and difficult to manage. The proposed system overcomes these limitations by providing a centralized platform for managing students, companies, job applications, and placement records efficiently.

The application is developed using Salesforce technologies such as Custom Objects, Apex Programming, Lightning Web Components (LWC), Salesforce Object Query Language (SOQL), Reports, and Dashboards. These technologies provide a secure and scalable environment for data management, business logic implementation, and real-time reporting. The system consists of four main modules: Student Management, Company Management, Application Management, and Placement Management, each designed to streamline different stages of the placement process.

The Placement Tracker System enables placement officers to maintain accurate records, monitor student applications, manage recruitment activities, and generate reports for analysis. Interactive dashboards provide graphical representations of placement statistics, helping users make informed decisions quickly and efficiently. The use of Salesforce's cloud platform eliminates the need for maintaining dedicated infrastructure while ensuring high availability, data security, and easy accessibility.

The developed system significantly reduces manual effort, improves data accuracy, and enhances the overall efficiency of campus placement management. Its modular design also allows future enhancements such as email notifications, mobile application support, and integration with online assessment platforms. Overall, the Placement Tracker System demonstrates the effective use of Salesforce CRM in developing a secure, scalable, and user-friendly solution for automating placement activities in educational institutions.

CHAPTER 195

INTRODUCTION

1.1 INTRODUCTION

In today's rapidly evolving digital world, educational institutions are increasingly adopting modern technologies to automate and simplify their administrative processes. One of the most important activities performed by colleges and universities is the campus placement process. The placement process involves managing student information, company details, job opportunities, applications, interviews, and final placement records. As the number of students and recruiting companies continues to grow, managing these activities manually becomes time-consuming, inefficient, and prone to errors.

Traditionally, placement cells maintained student records using paper files, spreadsheets, or standalone software applications. These methods often resulted in data redundancy, inconsistent information, difficulty in tracking applications, delayed report generation, and poor communication among stakeholders. Manual systems also make it difficult to generate analytical reports that help placement officers evaluate placement performance and make informed decisions.

To overcome these challenges, organizations and educational institutions are increasingly adopting cloud-based Customer Relationship Management (CRM) platforms. Salesforce CRM is one of the world's leading cloud computing platforms that enables organizations to efficiently manage data, automate workflows, develop custom applications, and generate real-time reports and dashboards. Salesforce provides a secure, scalable, and highly customizable environment for developing enterprise-level applications without the need to manage physical servers or complex infrastructure.

The **Placement Tracker System Using Salesforce CRM** is a cloud-based application developed to automate and simplify the campus placement process. The system provides a centralized platform where placement officers can manage student information, company records, job applications, and placement details efficiently. It also provides analytical reports and interactive dashboards that enable administrators to monitor placement statistics, track student progress, and evaluate placement performance.

The project has been developed using Salesforce technologies such as Custom Objects, Apex Programming, Lightning Web Components (LWC), SOQL, Validation Rules, Formula Fields, Reports, and Dashboards. These technologies provide a robust framework for building secure and scalable applications while minimizing manual effort.

The Placement Tracker System improves data accuracy, reduces paperwork, increases operational efficiency, and enables quick access to information from anywhere using Salesforce's cloud infrastructure. By automating the placement process, the system helps placement officers focus on improving placement opportunities rather than managing administrative tasks.

The project also demonstrates the practical implementation of Salesforce CRM concepts in solving real-world business problems. It integrates multiple Salesforce features into a single application, providing an efficient and user-friendly solution for educational institutions. The system is designed to be scalable, allowing additional features and modules to be incorporated in the future based on institutional requirements.

Overall, the Placement Tracker System represents a modern cloud-based solution for managing campus placement activities. It enhances productivity, ensures secure data management, provides real-time analytics, and supports better decision-making through interactive reports and dashboards.

1.2 PROJECT OVERVIEW

The **Placement Tracker System Using Salesforce CRM** is a cloud-based application developed to simplify and automate the campus placement process in educational institutions. The primary objective of the system is to provide a centralized platform for managing placement-related activities such as maintaining student records, company information, job applications, and placement details. By replacing manual record-keeping with an integrated digital solution, the system enhances operational efficiency, improves data accuracy, and supports better decision-making.

The application has been developed using the Salesforce Platform, one of the leading cloud-based Customer Relationship Management (CRM) platforms. Salesforce offers a secure, scalable, and highly customizable environment for developing enterprise applications. By utilizing Salesforce's cloud infrastructure, the Placement Tracker System enables authorized users to access and manage

placement information anytime and from any location without requiring additional hardware or complex software installations.

The Placement Tracker System consists of four major functional modules: Student Management, Company Management, Application Management, and Placement Management. These modules work together to streamline the entire placement lifecycle, beginning with student registration and ending with the final placement process.

The Student Management Module maintains detailed information about students participating in campus placements. It stores academic and personal details such as Student ID, Name, Email Address, Phone Number, Branch, CGPA, Skills, and Placement Status. The module allows placement officers to search, update, and manage student records efficiently while ensuring that student information remains organized and easily accessible.

The Company Management Module is responsible for maintaining information about organizations participating in campus recruitment. It stores details such as Company ID, Company Name, Package Offered, Location, HR Contact Information, Eligibility Criteria, Required Skills, and Last Application Date. This module enables placement officers to monitor recruiting companies and manage recruitment opportunities effectively.

The Application Management Module records applications submitted by students for various job opportunities. It maintains information regarding the student, company, job role, application status, and application date. The module allows placement officers to track the progress of every application throughout the recruitment process, ensuring that no application is overlooked or lost.

The Placement Management Module manages the final stage of the recruitment process by recording placement details for selected students. It maintains information such as Placement ID, Placement Number, Student Name, Company Name, Job Role, Package Offered, Joining Date, and Placement Status. This module helps generate placement statistics and provides valuable insights into the overall placement performance of the institution.

A significant feature of the Placement Tracker System is its Lightning Web Component (LWC) Dashboard, which provides a modern and interactive user interface. The dashboard displays

important Key Performance Indicators (KPIs), including Total Students, Total Companies, Total Applications, and Total Placements. It also includes search functionality and a student records table, enabling users to quickly locate and manage information.

In addition to the custom dashboard, the system makes extensive use of Salesforce Reports and Dashboards for data analysis and visualization. Reports such as Student Report, Company Report, Application Report, and Placement Report help users organize and analyze placement data. Dashboard components, including charts, graphs, and metrics, provide real-time insights into branch-wise student distribution, application status, company-wise placements, and overall placement performance.

The project incorporates several Salesforce technologies, including Custom Objects, Relationships, Validation Rules, Formula Fields, Apex Classes, SOQL Queries, Lightning Web Components (LWC), Reports, and Dashboards. These technologies collectively ensure that the application is secure, efficient, scalable, and capable of handling future enhancements.

Overall, the Placement Tracker System serves as a comprehensive solution for educational institutions seeking to automate their placement process. By integrating cloud computing with Salesforce CRM, the application minimizes manual effort, improves record management, enhances reporting capabilities, and supports efficient campus placement activities.

1.3 PROBLEM STATEMENT

Campus placement is one of the most significant activities carried out by educational institutions, as it plays a vital role in shaping students' careers and enhancing the institution's reputation. However, many colleges still rely on manual or partially computerized methods to manage placement-related activities. Student information, company details, job opportunities, applications, and placement records are often maintained using paper documents, spreadsheets, or multiple independent software applications. As the number of students and recruiting companies increases every academic year, managing these records manually becomes increasingly difficult and inefficient.

One of the major challenges of the existing system is the absence of a centralized platform for storing and managing placement data. Student information, company records, and application details are

often maintained separately, making it difficult to access complete and up-to-date information when required. This fragmented approach leads to duplication of records, inconsistencies in data, and increased chances of human error. As a result, placement officers spend a significant amount of time searching, verifying, and updating information instead of focusing on placement activities.

Another important issue is the difficulty in tracking the progress of student applications throughout the recruitment process. Since application details are maintained manually, monitoring whether a student has applied, been shortlisted, attended interviews, or secured placement becomes a time-consuming task. The absence of real-time tracking also makes it challenging to identify eligible candidates for upcoming recruitment drives and to monitor the overall placement performance of the institution.

Generating reports and analyzing placement data is another major challenge in traditional placement management systems. Preparing reports such as the total number of students placed, company-wise placements, branch-wise placement statistics, or application status often requires manual compilation of data from multiple sources. This process is not only time-consuming but also increases the possibility of errors, reducing the reliability of the information used for decision-making.

Data security and accessibility are also significant concerns in conventional systems. Paper records and locally stored files are vulnerable to loss, damage, unauthorized access, and accidental modification. Furthermore, users cannot easily access placement information from different locations, limiting collaboration among placement officers and administrators.

Considering these challenges, there is a need for a secure, centralized, and cloud-based placement management system capable of efficiently managing students, companies, job applications, and placement records. The system should automate routine tasks, maintain data accuracy, provide real-time information, simplify report generation, and support informed decision-making through analytical dashboards. The Placement Tracker System Using Salesforce CRM has been developed to address these challenges by providing an integrated platform that automates placement management, improves operational efficiency, and enhances the overall placement process for educational institutions.

1.4 OBJECTIVES OF THE PROJECT

The primary objective of the **Placement Tracker System Using Salesforce CRM** is to develop a cloud-based application that automates and simplifies the campus placement process. The system is designed to improve the efficiency of placement activities by providing a centralized platform for managing students, companies, job applications, and placement records.

The specific objectives of the project are as follows:

- To develop a centralized placement management system using Salesforce CRM.
- To maintain comprehensive records of students, including their academic and personal information.
- To manage company details such as eligibility criteria, package offered, location, HR contact information, and recruitment schedules.
- To provide an efficient mechanism for recording and tracking student job applications.
- To maintain placement records, including company name, job role, package, joining date, and placement status.
- To automate placement-related processes and reduce manual paperwork.
- To improve the accuracy, consistency, and security of placement data.
- To enable quick retrieval, updating, and management of placement information.
- To generate real-time reports for students, companies, applications, and placements.
- To provide analytical dashboards for monitoring placement performance and supporting decision-making.
- To develop an interactive and user-friendly interface using Lightning Web Components (LWC).

- To implement Apex programming, SOQL, Validation Rules, Formula Fields, and other Salesforce technologies in a real-world application.
- To develop a scalable and flexible system that can accommodate future enhancements and organizational requirements.

1.5 SCOPE OF THE PROJECT

The **Placement Tracker System Using Salesforce CRM** is developed to automate and simplify the campus placement process by providing a centralized and cloud-based platform for managing placement-related activities. The project focuses on improving the efficiency of placement operations by maintaining accurate records of students, companies, job applications, and placement details. It enables placement officers to perform various placement activities through a single integrated application while ensuring data security, consistency, and easy accessibility.

The scope of the project includes the following:

- To maintain complete academic and personal information of students participating in campus placements.
- To store and manage company information, including company profile, eligibility criteria, package offered, job location, required skills, HR contact details, and application deadlines.
- To facilitate the management and tracking of student job applications throughout the recruitment process.
- To maintain placement records, including company name, job role, offered package, joining date, and placement status.
- To provide a centralized repository for storing and retrieving placement-related information.
- To implement secure data management using Salesforce's cloud-based platform.
- To establish relationships among students, companies, applications, and placement records for efficient data management.

- To automate business processes using Salesforce features such as Validation Rules, Formula Fields, Apex Programming, and Lightning Web Components (LWC).
- To generate reports for students, companies, applications, and placements using Salesforce Reports.
- To provide interactive dashboards that display placement statistics and support data-driven decision-making.
- To provide a responsive and user-friendly interface that simplifies placement management activities.
- To minimize manual paperwork and improve the overall efficiency, accuracy, and reliability of placement operations.
- To develop a scalable application that can be extended with additional functionalities such as interview scheduling, notification services, online assessments, alumni management, and integration with external recruitment platforms in the future.

1.6 TECHNOLOGIES USED

The Placement Tracker System has been developed using Salesforce CRM and its associated technologies to provide a secure, scalable, and cloud-based solution for managing campus placement activities. The following technologies and tools were used during the development of the project.

Table 1.1 Technologies Used in the Project

Technology / Tool	Purpose
Salesforce CRM	Cloud-based platform used for developing the Placement Tracker System.
Apex	Server-side programming language used to implement business logic and controller classes.
SOQL (Salesforce Object Query Language)	Used to retrieve, filter, and manipulate records stored in Salesforce objects.
Lightning Web Components (LWC)	Used to develop an interactive and responsive user interface for the application.
HTML	Used to define the structure of Lightning Web Components.
CSS	Used to design and style the user interface of Lightning Web Components.
JavaScript	Used to implement client-side logic and dynamic functionality in Lightning Web Components.
Salesforce CLI (SFDX)	Used for project creation, deployment, and source management.
Visual Studio Code	Integrated Development Environment (IDE) used for writing and managing project code.

Custom Objects	Used to store data related to students, companies, applications, and placements.
Object Relationships	Used to establish logical associations among Salesforce objects.
Validation Rules	Used to validate user input and maintain data integrity.
Formula Fields	Used to perform automatic calculations based on field values.
Reports	Used to generate structured reports for students, companies, applications, and placements.
Dashboards	Used to visualize placement statistics through charts and graphical components.

1.7 REPORT ORGANIZATION

The report is organized into six chapters, each focusing on a specific aspect of the project. Chapter 1 introduces the Placement Tracker System and discusses the project overview, problem statement, objectives, scope, and technologies used. Chapter 2 presents the literature review and technical aspects related to Salesforce CRM and placement management systems. Chapter 3 discusses the functional and non-functional requirements, along with the hardware and software requirements of the project. Chapter 4 describes the technologies used for developing the application, including Salesforce CRM, Apex, SOQL, Lightning Web Components, Reports, and Dashboards. Chapter 5 explains the implementation of the project with code snippets, screenshots, and detailed descriptions of various modules. Finally, Chapter 6 presents the conclusion, highlights the achievements of the project, and discusses possible future enhancements.

CHAPTER 2

LITERATURE REVIEW / TECHNICAL ASPECTS

2.1 INTRODUCTION

A literature review is an important part of any software development project as it provides an understanding of existing systems, technologies, and research related to the proposed application. It helps in identifying the strengths and limitations of current solutions while establishing the need for developing an improved system. The review also provides a foundation for understanding the technical approaches and concepts relevant to the development of the proposed system.

Campus placement is one of the most significant activities in educational institutions, as it provides a connection between students and potential employers. The placement process involves several activities, including maintaining student information, managing company details, monitoring job applications, coordinating recruitment activities, maintaining placement records, and preparing placement-related reports. As the number of students and recruiting organizations increases, managing these activities manually becomes more difficult and time-consuming.

Traditional placement processes generally depend on paper records, spreadsheets, emails, and other disconnected methods of information management. Such approaches can lead to difficulties in maintaining updated records, tracking applications, preparing reports, and accessing information efficiently. These challenges have encouraged educational institutions to adopt computerized and web-based placement management systems that can reduce manual effort and improve the overall efficiency of placement activities.

The development of cloud-based technologies has further provided new possibilities for placement management. Cloud platforms can provide centralized data management, accessibility, scalability, automation, and reporting capabilities within a single environment. These features can help institutions manage placement information more efficiently while reducing their dependence on local infrastructure and manual processes.

The proposed Placement Tracker System Using Salesforce CRM has been developed to address these challenges by providing a centralized, cloud-based platform for managing placement-related information. The system manages important entities such as students, companies, applications, and placement records and provides reporting and dashboard capabilities for monitoring placement activities.

This chapter presents a review of traditional placement systems, existing placement management solutions, previous research, and the concept of Customer Relationship Management (CRM). It also discusses the research gap identified from existing approaches and provides a comparative analysis of traditional, web-based, and cloud-based placement management systems. The findings of this chapter provide the foundation and motivation for the development of the proposed Salesforce-based Placement Tracker System.

2.2 TRADITIONAL PLACEMENT MANAGEMENT SYSTEM

Traditionally, placement activities in colleges and universities were managed manually using paper records, registers, files, and spreadsheet applications such as Microsoft Excel. Placement officers maintained separate records for student information, company details, job applications, interview activities, and final placement records. Students generally submitted their resumes and application forms manually, while information about recruiting companies and available opportunities was recorded using physical documents or spreadsheets.

Although this approach was suitable for institutions managing a relatively small number of students and companies, it became increasingly inefficient as the scale of placement activities expanded. Placement officers had to perform several repetitive tasks, including entering student information, verifying eligibility criteria, updating application status, maintaining company records, and preparing placement statistics. These activities required considerable time and administrative effort.

One of the major limitations of the traditional system was the absence of centralized data management. Student, company, application, and placement information was often maintained in separate files or registers. As a result, retrieving complete and accurate information required searching through multiple sources.

Maintaining multiple files also increased the possibility of data duplication, missing records, and calculation errors. The same student or company information might have to be entered repeatedly in different documents. If an update was made in one file but not in another, inconsistencies could occur. Similarly, manually calculating placement statistics could result in errors and consume additional time.

Another significant limitation was the difficulty involved in tracking applications and recruitment progress. Placement officers had to manually update records whenever a student applied for a company, was shortlisted, attended an interview, or received a placement offer. As the number of applications increased, maintaining updated records became increasingly challenging.

Traditional systems also provided limited reporting and analytical capabilities. Preparing reports such as total placements, company-wise placements, branch-wise student information, and application statistics required placement officers to collect and compile information manually from different records. This delayed the availability of important placement information and made quick decision-making difficult.

Security was another concern associated with traditional placement management. Paper records could be misplaced, damaged, or accessed by unauthorized individuals. Similarly, locally stored spreadsheet files could be affected by accidental deletion, hardware failures, or unauthorized access. Maintaining reliable backups also required additional manual effort and appropriate storage arrangements.

The traditional approach also had limitations in terms of accessibility and collaboration. Placement information was generally available only on the computers or physical locations where the records were maintained. Placement officers and administrators could not easily access updated information from different locations. This restricted collaboration and made it difficult to share placement information efficiently.

Despite these limitations, traditional placement management systems established the fundamental workflow followed by modern placement applications. The major activities included student registration, company registration, application processing, eligibility verification, recruitment

tracking, and placement record maintenance. These basic processes later became the foundation for the development of computerized, web-based, and cloud-based placement management systems.

Therefore, the limitations of traditional placement management created a need for a more centralized and automated approach. The proposed Placement Tracker System Using Salesforce CRM addresses these challenges by providing centralized data management, automated processing, application tracking, reporting, dashboards, and cloud-based accessibility within a single platform.

2.3 EXISTING PLACEMENT MANAGEMENT SYSTEMS

With the advancement of web technologies, educational institutions have increasingly adopted computerized placement management systems to improve efficiency and reduce the manual effort involved in campus recruitment activities. These systems provide a digital environment for managing different stages of the placement process and offer better organization of placement-related information compared with traditional paper-based methods.

Most existing placement management systems are developed as web-based applications and provide functionalities such as student registration, company management, job application tracking, placement record maintenance, and report generation. These features allow institutions to maintain placement information in a structured digital format and reduce dependency on physical records and spreadsheets.

Such systems generally provide students with an online interface for maintaining their profiles, entering academic information, uploading resumes, viewing available job opportunities, and submitting applications. Students can also track the progress of their applications and obtain information about recruitment activities through the system. This improves accessibility and reduces the need for students to depend entirely on placement officers for routine activities.

Placement officers and administrators can use administrative interfaces to manage student information, company details, recruitment drives, eligibility requirements, applications, and placement records. Administrative dashboards can provide an overview of ongoing recruitment activities and help officers monitor the progress of different placement drives.

Many existing systems also provide basic reporting capabilities. Placement officers can generate reports containing information about registered students, participating companies, applications, and successful placements. These reports can help institutions prepare placement statistics and evaluate recruitment outcomes.

However, existing placement management systems still have several limitations. Many web-based applications require dedicated servers, databases, regular software maintenance, and technical administration. The institution may therefore need to manage infrastructure, backups, security configurations, and software updates. These requirements can increase the complexity and cost of maintaining the system.

Scalability is another important limitation. As the number of students, companies, applications, and placement records increases, systems with limited infrastructure may experience difficulties in handling the growing volume of data and users. A placement management system should therefore be capable of supporting increasing requirements without significantly affecting its performance or usability.

Another common limitation is the lack of advanced analytics and automation. Although basic reports may be available, interactive dashboards, automated workflows, and real-time monitoring capabilities may be limited in some existing systems. This can require placement officers to perform additional manual work when monitoring applications or preparing detailed placement statistics.

Security and backup management can also depend on the infrastructure and technical resources available to the organization. Maintaining secure access to student and company information requires appropriate authentication, authorization, backup, and data protection mechanisms. Institutions using locally maintained systems may need additional technical resources to manage these requirements.

These limitations have encouraged the adoption of cloud-based platforms for placement management. Cloud-based solutions can provide centralized data storage, improved accessibility, scalability, automation, security features, and reporting capabilities without requiring institutions to independently manage the complete underlying infrastructure.

Therefore, while existing placement management systems have significantly improved the efficiency

of campus recruitment compared with traditional methods, there remains a need for an integrated solution that combines centralized data management, automation, scalability, security, and interactive analytics. These requirements form the basis for the proposed Placement Tracker System using Salesforce CRM.

2.4 CUSTOMER RELATIONSHIP MANAGEMENT (CRM)

Customer Relationship Management (CRM) is a business strategy and software solution used to manage interactions between an organization and its customers or stakeholders. CRM systems provide a centralized platform for storing information, managing communication, automating business processes, and improving organizational efficiency. They enable organizations to maintain accurate records, monitor activities, and make informed decisions based on real-time information.

Modern CRM platforms integrate various business functions such as sales, marketing, customer service, reporting, and workflow automation into a single system. By maintaining all information within a centralized database, CRM systems reduce data redundancy, improve communication, and enhance productivity.

In the education sector, CRM platforms are increasingly used to manage student information, admissions, alumni relations, and placement activities. They help institutions maintain organized records, automate administrative processes, and generate reports for decision-making. Among the various CRM platforms available today, Salesforce CRM is one of the most widely adopted cloud-based solutions due to its scalability, security, and extensive customization capabilities.

The proposed Placement Tracker System utilizes Salesforce CRM to automate placement management activities. By integrating student records, company information, job applications, and placement details within a single platform, the system provides improved efficiency, better data management, and real-time reporting capabilities compared to traditional placement management systems.

2.5 CHALLENGES IN EXISTING PLACEMENT MANAGEMENT SYSTEMS

Although existing placement management systems have significantly improved the recruitment process compared to traditional manual methods, several challenges still exist. One of the major challenges is the lack of a centralized platform for managing all placement-related activities. In many institutions, student information, company details, applications, and placement records are maintained in separate systems, making data management difficult and increasing the chances of inconsistency.

Another challenge is limited automation. Many existing systems require manual verification of student eligibility, updating of application status, and preparation of placement reports. These manual activities consume time and increase the possibility of human errors.

Scalability is also a concern. As the number of students, companies, and placement records increases, conventional systems may experience performance issues. Institutions often need additional infrastructure and maintenance to support growing data volumes.

Data security is another important challenge. Student and company information is confidential and must be protected from unauthorized access. Traditional applications often require institutions to implement their own backup and security mechanisms, which increases maintenance efforts and operational costs.

Finally, many existing systems provide only basic reporting features. They lack interactive dashboards and real-time analytics that help placement officers monitor placement activities, identify trends, and make informed decisions.

2.6 REVIEW OF EXISTING RESEARCH

Several researchers have proposed different approaches for automating campus placement activities using web technologies, database management systems, and cloud computing. Most of these studies focus on reducing manual work, improving data organization, and providing students and placement officers with easier access to placement information.

Web-based placement systems commonly provide features such as online student registration, resume uploading, job application, application status tracking, and company information management. Administrative dashboards also help placement officers monitor recruitment activities and maintain placement records. Although such systems improve efficiency compared with traditional manual methods, some solutions may have limitations in scalability, infrastructure management, and advanced analytical capabilities.

Recent research has increasingly considered cloud-based applications because of their flexibility, accessibility, centralized data management, and reduced infrastructure requirements. Cloud platforms can also support automatic updates, improved availability, and easier expansion as the number of users and records increases. Research has further emphasized the importance of integrating reporting and analytical capabilities to support effective monitoring and decision-making.

The findings from existing research indicate that an effective placement management system should provide centralized data storage, secure access, workflow automation, scalability, and reporting capabilities. These observations provide the basis for the development of the proposed Placement Tracker System using Salesforce CRM, which combines cloud-based data management, automation, application tracking, and analytical dashboards within a centralized platform.

2.7 COMPARATIVE ANALYSIS

A comparative analysis helps evaluate the strengths and limitations of different approaches used for managing campus placement activities. Over time, educational institutions have gradually moved from traditional manual methods to web-based applications and, more recently, to cloud-based platforms such as Customer Relationship Management (CRM) systems. Each approach differs in terms of data management, accessibility, security, automation, scalability, maintenance, and reporting capabilities.

Traditional placement management systems mainly depend on paper records, spreadsheets, and manual documentation. Although these methods are simple to implement, they become difficult to manage when the number of students, companies, applications, and placement records increases. Manual data entry can also result in duplication, inconsistency, and human errors.

Web-based placement management systems provide improvements by converting physical records into digital records and allowing users to access information through a web interface. These systems can provide features such as student registration, company management, application tracking, and basic reporting. However, depending on their architecture, they may require dedicated servers, databases, maintenance, security configuration, and regular software updates. Their scalability and analytical capabilities may also vary depending on the technologies used for implementation.

Cloud-based CRM platforms provide a more integrated approach to placement management. Salesforce, for example, combines centralized data management with application development, automation, security, reporting, and dashboard capabilities. Since the platform is cloud-based, users can access the application through an internet connection without requiring the institution to maintain separate application and database infrastructure.

Cloud-based systems also provide better opportunities for automation of routine placement activities. Processes such as updating records, validating information, tracking applications, and generating reports can be supported through platform-based automation. This can reduce repetitive administrative work and allow placement officers to focus more on monitoring recruitment activities and assisting students.

Another important difference is the analytical capability provided by the different approaches. Traditional systems generally require users to manually prepare summaries and placement statistics. Web-based systems can provide digital reports, but the level of analysis depends on the features implemented in the application. Salesforce provides built-in Reports and Dashboards that can present placement information through charts, tables, and key performance indicators, making it easier to understand recruitment trends.

Data relationships are another important consideration. In traditional systems, information about students, companies, applications, and placements may be maintained in separate files. Web-based applications can establish relationships between database tables, but this requires appropriate database design and implementation. Salesforce allows related records to be connected through object relationships, enabling information from different placement modules to be accessed together.

The approaches also differ in terms of maintenance and infrastructure requirements. Traditional systems require continuous manual maintenance of records, while web-based systems may require administrators to manage application servers, databases, backups, and software updates. In a cloud-based Salesforce environment, much of the underlying infrastructure is managed by the platform, allowing developers and administrators to focus primarily on application configuration and functionality.

User accessibility is another important factor in placement management. Traditional records are generally accessible only at the location where they are maintained. Web-based systems improve accessibility through network or internet access, while cloud-based systems provide access through the Salesforce platform, subject to appropriate permissions and connectivity.

The comparison also highlights the importance of data security and controlled access. Manual records can be difficult to protect and monitor, while security in web-based systems depends largely on how the application is designed and maintained. Salesforce provides platform-level security and access-control mechanisms that can be configured according to organizational requirements.

Considering these factors, cloud-based CRM platforms provide a more comprehensive approach to placement management by combining centralized data management, automation, accessibility, security, scalability, reporting, and analytics. For the proposed Placement Tracker System, Salesforce CRM therefore provides a suitable foundation for integrating different placement activities into a single environment.

Overall, the comparative analysis indicates that the transition from traditional systems to web-based and cloud-based solutions can significantly improve the organization and management of placement activities. The proposed Salesforce-based system builds upon these advantages by combining operational placement management with reporting and dashboard-based analytics within the same platform.

Table 2.1 Comparative Analysis of Placement Management Systems

Feature	Traditional System	Web-Based System	Salesforce-Based System
Data Storage	Paper Files / Excel	Database	Cloud Database
Accessibility	Limited	Internet	Anywhere, Anytime
Security	Low	Medium	High
Scalability	Low	Medium	High
Automation	Manual	Partial	Extensive
Report Generation	Manual	Basic	Advanced
Dashboard Support	Not Available	Limited	Interactive
Maintenance	High	Moderate	Low
Backup	Manual	Organization Managed	Automatic
Data Sharing	Difficult	Moderate	Easy and Secure

2.8 RESEARCH GAP

The literature review indicates that although numerous placement management systems have been developed to support educational institutions in managing student placement activities, many existing solutions continue to face several limitations. Traditional placement processes are often dependent on manual record keeping, spreadsheets, emails, and physical documentation. Such approaches can become difficult to manage when the number of students, companies, applications, interviews, and placement records increases. The dependency on manual processes may also result in duplication of information, delayed updates, difficulty in tracking application status, and increased administrative workload.

Another important limitation identified from the review is the fragmentation of placement-related data. In many traditional and basic web-based systems, student information, company details, job opportunities, applications, interviews, and placement records may be maintained separately. When information is distributed across multiple files or systems, it becomes difficult for placement coordinators to obtain a complete view of the placement process. For example, information about a student's eligibility may be stored separately from information about the companies for which the student has applied. Similarly, application status and final placement information may not always be connected effectively. This fragmentation reduces the efficiency of information retrieval and makes overall placement monitoring more difficult.

The literature also highlights limitations related to automation. Traditional placement systems frequently require placement coordinators to manually update application statuses, maintain student records, track company opportunities, and prepare placement reports. Such activities consume considerable time, particularly during active recruitment periods when many companies and students are involved simultaneously. A modern placement management system should therefore reduce repetitive administrative activities by providing automated workflows and centralized data management.

A further research gap is related to scalability. Many basic placement applications are designed for a limited number of users or depend on local infrastructure. While such systems may work effectively for small-scale demonstrations or limited institutional requirements, their management becomes

more challenging as the volume of data and number of users increases. A scalable system should be capable of managing increasing numbers of student records, company records, applications, and placement activities without requiring major changes to the underlying infrastructure.

Another limitation identified in existing solutions is the lack of comprehensive analytical capabilities. Several systems provide basic facilities for storing and retrieving placement records but offer limited options for analysing the collected information. Simply storing data does not provide sufficient support for institutional decision-making. Placement coordinators and management may require information such as the number of students placed, branch-wise placement distribution, company-wise recruitment, application status, and overall placement trends. Interactive reports and dashboards can transform stored records into useful information that can support monitoring and decision-making.

The review also indicates a need for improved centralized data management. Placement activities involve multiple types of information, including student profiles, academic information, company details, job roles, application records, placement information, and communication-related details. Managing these records within a centralized environment can reduce data duplication and improve consistency. It can also make it easier for authorized users to access the latest information from a common system rather than depending on multiple disconnected sources.

Security and controlled access are also important considerations in placement management. Student and company records may contain sensitive information such as contact details, academic performance, application information, and recruitment-related records. Therefore, a placement management solution should provide an environment in which information can be stored and accessed in a controlled manner. Different users may require different levels of access depending on their responsibilities. For example, a placement administrator may need access to a broader set of records than an individual student. This creates a requirement for a platform capable of supporting centralized data management together with appropriate access control.

Another significant gap is the limited integration of workflow management and reporting within the same environment. In many systems, data management, process automation, and reporting may exist as separate functions. This can require users to switch between different tools for maintaining

records, tracking processes, and generating reports. An integrated platform can reduce this fragmentation by bringing these capabilities together.

The literature review further demonstrates that conventional systems may not provide sufficient support for real-time monitoring. Placement activities can change frequently. A student's application status may change from applied to shortlisted, interviewed, selected, or rejected. Similarly, new companies may be added and placement records may be updated throughout the recruitment process. If reports are prepared manually, the information may not always represent the latest situation. A centralized system with updated records and interactive reporting can provide more timely information to placement coordinators and management.

The need for cloud-based placement management is therefore another important research gap. Traditional applications may require local installation, maintenance, and infrastructure management. A cloud-based approach can provide centralized access to application data and reduce dependency on a particular local environment. It can also provide a foundation for scalability and accessibility while supporting organizational requirements for centralized data management.

Another gap identified is the limited use of enterprise-level CRM platforms for placement management. Customer Relationship Management platforms are traditionally associated with managing relationships, records, workflows, and interactions in business environments. However, the concepts of centralized information management, workflow automation, reporting, and relationship management can also be applied to educational placement processes. Students can be managed as placement participants, companies can be maintained as recruitment partners, and applications and placements can be tracked as related records within the same environment.

The proposed Placement Tracker System Using Salesforce CRM addresses these identified research gaps by utilizing Salesforce as a cloud-based platform for centralized placement management. The system is designed to manage important placement entities such as student information, company details, applications, and placement records within an integrated environment. Relationships between these entities allow relevant information to be connected and accessed more efficiently.

The proposed system also addresses the need for automation and structured data management through Salesforce capabilities. Apex-based functionality, Salesforce objects, relationships, validation mechanisms, and workflows can be used to support the management of placement-related processes.

The system further addresses the analytical gap through the implementation of reports and dashboards. The proposed solution provides an interactive Lightning Web Component dashboard containing important placement information such as total students, total companies, total applications, and total placements. In addition, Salesforce Reports and Dashboards can be used to analyse information such as students by branch, applications by status, placements by company, and overall placement counts. These analytical features provide users with a clearer view of the current placement situation.

The use of Salesforce CRM also provides a foundation for scalability and centralized management. Instead of maintaining multiple disconnected files, placement-related information can be maintained within a common cloud-based environment. This approach can support the future expansion of the system as the number of students, companies, applications, and placement records increases.

The proposed system therefore attempts to bridge the gap between basic placement record-management applications and a more integrated enterprise-level placement management solution. Rather than focusing only on storing placement data, the system combines centralized data storage, relationships, validation, automation, application tracking, reporting, dashboards, and cloud-based accessibility within a single platform.

Overall, the research gap identified from the literature can be summarized as the need for a placement management system that is centralized, scalable, automated, analytical, and cloud-based. Existing approaches provide useful solutions for individual aspects of placement management, but there remains a need for an integrated solution that combines these capabilities. The proposed Placement Tracker System Using Salesforce CRM is designed to address this requirement by providing a unified environment for managing student placement activities and generating meaningful information for monitoring and decision-making.

2.9 CHAPTER SUMMARY

This chapter presented a review of the concepts, existing approaches, and challenges related to the proposed Placement Tracker System Using Salesforce CRM. It discussed the traditional placement management process and highlighted the difficulties associated with manual records, spreadsheets, emails, and disconnected methods of managing placement information.

The chapter examined existing placement management systems and identified limitations such as limited automation, fragmented data management, scalability issues, inadequate reporting, and dependence on local infrastructure. The concept of Customer Relationship Management (CRM) was also discussed, highlighting how centralized data management, relationships, workflows, and reporting capabilities can be applied to educational placement activities.

The review emphasized the importance of a cloud-based placement management solution capable of maintaining student, company, application, and placement information within a centralized environment. Cloud-based systems can improve accessibility and scalability while reducing dependence on local infrastructure. The importance of reports and analytics was also highlighted, as placement data can provide useful information about student participation, company recruitment, application progress, and placement outcomes.

Based on the literature review, a research gap was identified in the lack of integrated solutions that combine centralized data management, automation, scalability, cloud-based infrastructure, and interactive analytics within a single platform. The proposed Placement Tracker System Using Salesforce CRM addresses this gap by using Salesforce to manage student information, company details, applications, and placement records in an integrated environment.

The proposed system incorporates Salesforce Objects, relationships, Apex programming, validation mechanisms, Lightning Web Components, Reports, and Dashboards. The customized LWC dashboard provides important information such as total students, companies, applications, and placements, while Salesforce Reports and Dashboards provide analytical insights such as students by branch, applications by status, and placements by company.

CHAPTER 3

REQUIREMENT ANALYSIS (S/W, H/W)

3.1 INTRODUCTION

Requirement analysis is one of the most important phases of the Software Development Life Cycle (SDLC) because it establishes the foundation for designing and developing a successful software system. It involves identifying, analyzing, organizing, and documenting the requirements of the proposed application. A proper requirement analysis helps in understanding the needs of users, defining system objectives, identifying technical constraints, and determining the resources required for implementation. It also reduces the possibility of errors during development and ensures that the final system satisfies the intended requirements.

For the Placement Tracker System Using Salesforce CRM, requirement analysis focuses on understanding the requirements involved in managing the complete placement process. The system is designed to manage important information related to students, companies, job applications, and placement records in a centralized environment. It also considers the requirements for maintaining relationships between these entities, tracking application and placement status, and providing useful information through reports and dashboards.

The requirements of the system can be broadly categorized into functional and non-functional requirements. Functional requirements describe the specific operations and services that the system should provide, such as creating and managing student and company records, tracking applications, maintaining placement information, and displaying placement statistics. Non-functional requirements define the expected quality and performance of the system, including security, scalability, usability, reliability, and accessibility.

The requirement analysis also identifies the software and hardware requirements necessary for system development and deployment. Since the proposed application is developed using the Salesforce cloud platform, major infrastructure components such as servers, database management,

and platform availability are provided by Salesforce. Therefore, developers primarily require a suitable computer system, internet connectivity, a web browser, Salesforce Developer Edition, and development tools such as Visual Studio Code and Salesforce CLI.

This chapter therefore defines the requirements necessary for developing the Placement Tracker System and provides a clear understanding of the resources, functionalities, and quality attributes expected from the proposed solution. The identified requirements serve as the foundation for the subsequent stages of system design, development, implementation, testing, and evaluation.

3.2 FUNCTIONAL REQUIREMENTS

Functional requirements describe the operations and services that the Placement Tracker System must perform to meet user requirements. The major functional requirements of the system are as follows:

- The system shall maintain student information, including personal and academic details.
- The system shall maintain company records, including eligibility criteria, package, location, HR details, and application deadlines.
- The system shall allow placement officers to manage student job applications.
- The system shall maintain placement details of selected students.
- The system shall provide search functionality for retrieving student, company, application, and placement records.
- The system shall generate reports related to students, companies, applications, and placements.
- The system shall provide dashboards for visualizing placement statistics.
- The system shall validate user input using validation rules.
- The system shall maintain relationships among different Salesforce objects.

3.3 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes and performance characteristics of the system. The Placement Tracker System should satisfy the following non-functional requirements:

- The system should provide a user-friendly interface.
- The application should be secure and accessible only to authorized users.
- The system should provide high availability through the Salesforce cloud platform.
- The application should support future enhancements without major modifications.
- The system should ensure data accuracy and consistency.
- The application should provide quick response time while retrieving records.
- The system should be scalable to handle increasing numbers of students and placement records.
- The system should maintain data integrity through validation and relationships.

3.4 SOFTWARE REQUIREMENTS

The Placement Tracker System was developed using Salesforce CRM as the primary cloud-based development platform, along with several supporting software tools required for coding, deployment, testing, debugging, and project management. Since Salesforce provides a complete cloud environment for application development, most of the database management, server-side infrastructure, security, and application hosting requirements are handled by the platform itself. The supporting software tools were mainly used to create, modify, test, deploy, and maintain the different components of the application.

Salesforce CRM served as the central platform for developing and managing the Placement Tracker System. Custom Objects were created to store information related to students, companies, applications, and placements. Salesforce also provided features such as object relationships,

validation rules, formula fields, reports, dashboards, and user access controls. These built-in features reduced the need to develop every functionality from the beginning.

Visual Studio Code was used as the primary source code editor during development. It provided an environment for writing and modifying Apex classes, Apex triggers, Lightning Web Components, JavaScript files, CSS files, HTML files, and XML configuration files. Salesforce Extensions for Visual Studio Code were used to provide Salesforce-specific development features and improve coding productivity.

Salesforce CLI, also referred to as Salesforce DX tools, was used for Salesforce project management and source-driven development. It supported activities such as creating the project structure, authorizing the Salesforce organization, retrieving metadata, deploying source code, and synchronizing project files with the Salesforce environment. The use of Salesforce CLI also helped maintain an organized development workflow.

A modern web browser such as Google Chrome or Microsoft Edge was required for accessing Salesforce and testing the developed application. The browser provided access to Salesforce Lightning Experience, Setup, Object Manager, Reports, Dashboards, and the custom Placement Tracker application. A stable internet connection was necessary because Salesforce is a cloud-based platform.

The development environment also required Java Development Kit support for certain Salesforce development tools and extensions. The required software components were configured on the development computer before beginning the implementation of the project. Proper configuration of these tools allowed the developer to work efficiently with Salesforce metadata, Apex code, Lightning Web Components, and deployment commands.

Table 3.1 Software Requirements

Software	Purpose
Windows 10/11	Operating System
Salesforce CRM	Application Development Platform
Visual Studio Code	Source Code Development
Salesforce CLI (SFDX)	Deployment and Project Management
Google Chrome / Microsoft Edge	Application Testing
Java Development Kit (JDK)	Salesforce Extension Support

3.5 HARDWARE REQUIREMENTS

The Placement Tracker System does not require high-end or specialized hardware because the application is developed and deployed using the Salesforce cloud platform. Unlike traditional applications that may require dedicated servers, database servers, and additional infrastructure, the proposed system uses Salesforce for application hosting, data storage, and server-side processing. Therefore, a standard computer system with adequate processing power, memory, storage, display, and internet connectivity is sufficient for development, testing, and regular access to the application.

The processor is responsible for executing development tools such as Visual Studio Code, Salesforce CLI, web browsers, and other supporting applications. An Intel Core i3 or AMD Ryzen 3 processor or above is considered sufficient for the development requirements of the project. A higher-

performance processor can provide smoother operation when multiple development applications are running simultaneously.

A minimum of 8 GB RAM is recommended for the development environment. Adequate memory is required because Visual Studio Code, browser applications, Salesforce tools, and other utilities may operate simultaneously during development and testing. Additional RAM can improve system responsiveness when working with larger projects or multiple applications.

Storage is required for installing the operating system, Visual Studio Code, Salesforce CLI, Java Development Kit, Salesforce extensions, project files, and other supporting software. At least 20 GB of free disk space is considered sufficient for the project development environment. Since the main application data is stored in Salesforce, extensive local storage is not required for maintaining the application's database.

A stable internet connection is one of the most important hardware and connectivity requirements for this project. Salesforce is a cloud-based platform, so users and developers need internet access to log in to the Salesforce organization, access records, execute queries, test applications, retrieve metadata, and deploy components. A reliable broadband connection helps prevent interruptions during development and testing.

A display resolution of 1366×768 or higher is recommended for comfortably working with Salesforce Lightning Experience, Visual Studio Code, browser-based development tools, and dashboard interfaces. A larger display can also be useful when working with code and Salesforce configuration screens simultaneously.

The hardware requirements are therefore relatively modest compared with applications that require local servers or database infrastructure. The major hardware requirements for the proposed system are presented in Table 3.2.

Table 3.2 Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 / AMD Ryzen 3 or above
RAM	Minimum 8 GB
Storage	20 GB Free Disk Space
Internet Connection	Broadband Connection
Display	1366 × 768 Resolution or Higher

3.6 FEASIBILITY ANALYSIS

Feasibility analysis is an important stage of the Software Development Life Cycle because it determines whether a proposed system can be successfully developed, implemented, and operated using the available resources. It is performed before or during the early stages of system development to identify possible difficulties and determine whether the proposed solution is practical. A proper feasibility study helps developers and organizations understand the technical requirements, expected benefits, operational requirements, and possible limitations of the proposed system.

For the Placement Tracker System, feasibility analysis is particularly important because the system is intended to replace or improve several activities that are traditionally performed using paper records, spreadsheets, emails, and manually maintained documents. The feasibility study evaluates whether these activities can be effectively transferred to a centralized Salesforce-based environment and whether the proposed system can provide meaningful improvements to the placement management process.

The feasibility analysis also helps determine whether the selected technologies are appropriate for the requirements of the project. The proposed system uses Salesforce CRM as the primary development platform, together with Apex, SOQL, Lightning Web Components, Custom Objects, Validation Rules, Reports, Dashboards, Salesforce CLI, and Visual Studio Code. These technologies provide the development environment required for implementing the major functions of the system.

The analysis considers the availability of hardware and software resources required for development and testing. Since Salesforce is a cloud-based platform, the system does not require a dedicated local application server or database server. A standard development computer with sufficient memory, storage, processing capability, and internet connectivity can be used to develop and test the application.

The feasibility study also considers the expected benefits of centralizing placement information. In a traditional environment, student details, company information, applications, and placement records may be distributed across several files. This can make information retrieval difficult and can increase the possibility of inconsistent or outdated records. The proposed system attempts to provide a single environment in which these records can be maintained and connected.

Another objective of the feasibility analysis is to determine whether placement officers and other authorized users can practically operate the system. The proposed application provides a customized dashboard and Salesforce interface that allow users to access placement information without requiring programming knowledge. The system is therefore evaluated not only from a technical perspective but also from the perspective of its everyday use.

The feasibility analysis for the Placement Tracker System is divided into three major areas: technical feasibility, economic feasibility, and operational feasibility. Technical feasibility evaluates the availability and suitability of the technologies. Economic feasibility evaluates the potential benefits and resource requirements. Operational feasibility evaluates the usability and suitability of the system for placement-related activities.

3.6.1 Technical Feasibility

Technical feasibility determines whether the proposed system can be developed successfully using the available technologies, software tools, hardware resources, and technical knowledge. A system is considered technically feasible when the required technologies are available and capable of supporting the expected functionality without creating unreasonable technical difficulties.

The Placement Tracker System is technically feasible because Salesforce CRM provides a cloud-based development environment containing most of the technologies required for the implementation of the proposed application. Salesforce supports both declarative and programmatic development approaches, allowing developers to combine configuration-based features with custom programming according to the requirements of the application.

The first major technical component of the system is the Salesforce data model. The proposed application uses Custom Objects to represent the main entities involved in campus placement management. The Student, Company, Application, and Placement objects provide a structured method of organizing placement-related information. Each object contains fields that represent the attributes required for managing the corresponding records.

For example, the Student object can contain information such as Student ID, Student Name, Email, Phone Number, Branch, CGPA, Skills, and Placement Status. Similarly, the Company object can contain Company Name, Package Offered, Job Location, Eligibility CGPA, Required Skills, HR Name, HR Email, and Last Application Date. This object-based structure makes it possible to organize information according to the requirements of the placement process.

The use of Object Relationships further supports the technical design of the system. The Student object is connected to Application and Placement records, while the Company object is also connected to Application and Placement records. These relationships allow related information to be retrieved without repeatedly storing the same student or company information in multiple locations.

This approach improves the organization of the data and supports more efficient reporting. For example, an application record can identify both the student who applied and the company for which the application was submitted.

Apex is another important technical component used in the proposed system. Apex is Salesforce's programming language for implementing customized server-side business logic. It allows the developer to create classes and methods that perform specific operations on Salesforce data. In the Placement Tracker System, Apex can be used to retrieve records, process information, perform required operations, and provide data to the Lightning Web Component.

Apex also supports object-oriented programming concepts such as classes, methods, collections, loops, conditional statements, and exception handling. These capabilities make it possible to develop structured and reusable business logic rather than placing all functionality directly inside the user interface.

SOQL provides the data retrieval mechanism required by Apex. The application can use SOQL queries to retrieve specific student, company, application, or placement records from Salesforce. Filtering conditions can be applied to retrieve only the required information. This is particularly useful for implementing search functionality, dashboard statistics, and other data-driven features.

For example, the dashboard may need to retrieve the total number of students or obtain student records according to a selected branch. SOQL can be used to retrieve the required records, which can then be processed by Apex and displayed through the Lightning Web Component.

Lightning Web Components provide the frontend development technology used for the customized dashboard. LWC follows a component-based architecture and uses standard web technologies such as HTML, CSS, and JavaScript. This allows the developer to create an interface that is more customized than a standard record-management screen.

The Placement Tracker Dashboard provides important placement information through KPI cards, search functionality, filtering options, and a student records table. These features demonstrate that the Salesforce platform is capable of supporting a customized user interface in addition to its standard administrative functionality.

Validation Rules provide another technically useful feature. They can be configured to ensure that users enter valid and consistent information before records are saved. This reduces the possibility of invalid data entering the system and improves overall data quality.

Reports and Dashboards provide the analytical layer of the proposed system. Reports can organize and summarize placement records, while dashboard components can convert these records into graphical and numerical representations. The system can therefore provide information such as students by branch, applications by status, placements by company, and overall placement statistics.

The development environment also contributes to technical feasibility. Visual Studio Code provides a suitable environment for creating Apex classes, LWC components, JavaScript files, CSS files, HTML files, and metadata. Salesforce Extensions provide additional functionality specifically for Salesforce development.

Salesforce CLI supports project creation, authentication, source retrieval, metadata deployment, and synchronization between the local development environment and the Salesforce organization. This makes the development process more structured and supports source-driven development.

The cloud-based nature of Salesforce is another important technical advantage. The institution does not need to independently configure and maintain an application server, database server, or other underlying infrastructure for the application. This reduces infrastructure-related complexity and allows developers to concentrate on application functionality.

The system can also be expanded as requirements change. Additional custom fields, objects, Apex classes, LWC components, reports, dashboards, and automation features can be added in future versions. This makes the platform suitable not only for the current project but also for possible future improvements.

Technical feasibility also depends on the availability of skilled users and developers. Developers need to understand Salesforce concepts, Apex, SOQL, LWC, metadata, and deployment procedures. However, the required technologies are available within the selected development environment, making the project technically achievable.

Security is another consideration. Salesforce provides mechanisms for controlling access to records and application functionality. Appropriate permissions can be configured for different users depending on institutional requirements. This is important because placement systems may contain student academic information and company recruitment information.

Scalability is also supported by the cloud-based architecture. As the number of placement records increases, the application can continue to use the Salesforce environment rather than requiring the institution to independently scale a local server infrastructure. The actual scalability available depends on the Salesforce environment and organizational requirements.

Considering the available platform capabilities, development tools, and technical resources, the Placement Tracker System is technically feasible. Salesforce provides the required environment for implementing the data model, business logic, user interface, reporting, analytics, and future extensions of the system.

3.6.2 Economic Feasibility

Economic feasibility determines whether the proposed system provides sufficient benefits compared with the resources and costs required for its development, implementation, and operation. It considers direct and indirect costs as well as improvements in productivity, efficiency, and resource utilization.

Traditional placement management can involve significant administrative costs. Placement departments may need to maintain paper records, registers, spreadsheets, printed forms, company documents, student resumes, and manually prepared reports. As the number of students and recruiting companies increases, maintaining these records can require additional time and administrative resources.

The proposed Placement Tracker System aims to reduce these requirements by providing a centralized digital environment. Student records, company details, application information, and placement outcomes can be maintained electronically. This reduces the dependency on physical documents and separate spreadsheets.

One of the major economic benefits of the system is the reduction in repetitive manual work. Placement officers may otherwise spend considerable time entering student information, updating application statuses, checking eligibility, searching records, and preparing placement statistics. By centralizing these activities, the system can reduce the time required for routine administrative operations.

The reduction in manual work can result in better utilization of staff time. Placement personnel can spend less time performing repetitive record-management activities and more time coordinating recruitment activities, communicating with companies, assisting students, and monitoring placement progress.

Paperwork is another area where cost reduction can be achieved. Physical documents require printing, storage, organization, and maintenance. Digital storage within Salesforce can reduce the need for maintaining large volumes of physical placement documentation for routine activities.

The proposed system can also reduce the effort required for report preparation. In a manual environment, placement officers may need to collect information from different spreadsheets and calculate statistics separately. Salesforce Reports can organize the information automatically based on the available records and filters.

Dashboards provide an additional economic benefit because they allow important statistics to be displayed visually. Instead of manually calculating the number of students, applications, or placements, users can obtain summarized information through KPI components and dashboard reports.

Another important consideration is data accuracy. Errors in manually maintained records can result in additional administrative work because incorrect information has to be identified and corrected. Centralized data management and validation mechanisms can help reduce such inconsistencies.

The cloud-based architecture can also reduce infrastructure-related requirements. Traditional systems may require dedicated application servers, database servers, backup systems, and technical staff for maintenance. The Salesforce platform provides the underlying cloud environment, reducing the need for the institution to independently maintain the complete infrastructure.

The system also provides value through improved accessibility. Authorized users can access the application through the Salesforce environment rather than depending on a particular computer containing a local database or spreadsheet. This can reduce operational limitations and support more flexible working practices.

From a development perspective, Salesforce also provides many built-in features that would otherwise need to be developed independently. For example, data storage, authentication, reporting, dashboards, and platform-level security capabilities are available as part of the Salesforce environment. This can reduce development effort for these common application requirements.

However, economic feasibility must also consider the cost of Salesforce licensing and services. The actual cost depends on the Salesforce edition, number of users, features required, storage requirements, and other organizational considerations. Therefore, an educational institution planning a large-scale deployment would need to evaluate the appropriate Salesforce licensing model.

The project should therefore not be considered completely cost-free. Although cloud infrastructure reduces the need for local hardware and maintenance, platform subscriptions or organizational Salesforce resources may still be required for production deployment.

For the purpose of the proposed academic project, the Salesforce environment provides an appropriate platform for demonstrating the feasibility of cloud-based placement management. The project focuses on demonstrating how centralized data, automation, reporting, and dashboards can reduce administrative effort.

The system can also provide long-term value if implemented at an institutional level. As placement activities grow, the value of centralized records and automated reporting can become more significant because manual management becomes increasingly time-consuming.

The economic benefits are therefore not limited to direct financial savings. They also include reduced administrative effort, improved productivity, reduced paperwork, faster information retrieval, improved reporting, and better utilization of placement personnel.

Considering these factors, the Placement Tracker System is economically feasible when implemented according to the institution's available Salesforce resources and licensing requirements. The potential benefits of reducing manual effort and improving placement management can provide significant operational value.

3.6.3 Operational Feasibility

Operational feasibility determines whether the proposed system can be successfully used in the actual working environment and whether it supports the requirements of its intended users. A technically successful system may still fail if users find it difficult to operate or if it does not fit into existing organizational processes. Therefore, operational feasibility is an important consideration for the Placement Tracker System.

The proposed system is designed primarily for placement officers, placement coordinators, administrators, and other authorized users involved in campus recruitment activities. Its main purpose is to simplify the management of students, companies, applications, and placement records.

The centralized structure of the system improves the overall placement workflow. Instead of maintaining separate spreadsheets for students, companies, applications, and placements, authorized users can access related information through Salesforce. This reduces the need to switch between different files and applications.

The Student module allows placement personnel to maintain student information in an organized manner. Details such as student ID, name, email, branch, CGPA, skills, and placement status can be maintained in the system. This allows placement officers to retrieve student information when required.

The Company module provides a centralized location for maintaining recruitment-related information. Company details, package information, job location, eligibility criteria, required skills, HR details, and application deadlines can be maintained in structured records.

The Application module helps placement officers track applications submitted by students. Each application can be connected to the relevant student and company. This makes it easier to monitor application status and recruitment progress.

The Placement module records the final placement information of students who successfully obtain employment. These records can be used for institutional reporting and placement analysis.

The LWC dashboard provides an important operational interface. Instead of requiring users to

navigate through multiple Salesforce objects to understand placement statistics, the dashboard provides an overview in a single interface.

The dashboard displays KPI cards representing the total number of students, companies, applications, and placements. These indicators provide a quick understanding of the current placement situation.

Search functionality improves the usability of the application by allowing users to locate student information quickly. Branch and placement-status filters can further narrow the displayed information and help users identify relevant records.

Reports provide another operational advantage. Placement officers can access structured reports for students, companies, applications, and placements. These reports can be filtered, sorted, grouped, and summarized according to the information requirements.

The Salesforce Dashboard provides visual analytics for management and placement coordinators. Information such as branch-wise student distribution, application status, and company-wise placement can be displayed through charts and other dashboard components.

This analytical capability can support decision-making. For example, placement coordinators can observe the number of applications submitted by students, monitor the progress of recruitment activities, and understand placement distribution across companies or branches.

The proposed system also reduces the amount of technical knowledge required from end users. Users do not need to understand Apex, SOQL, JavaScript, HTML, CSS, or LWC implementation to perform their daily activities. These technologies operate behind the user interface.

Basic training can be provided to placement officers covering Salesforce navigation, record creation, record updating, searching, filtering, and report access. Once users understand these basic operations, they can perform routine placement activities without requiring programming knowledge.

The system also supports better coordination among placement personnel. Since the information is maintained in a centralized environment, authorized users can work with the same set of records. This reduces the possibility of maintaining multiple conflicting versions of the same placement information.

Cloud-based accessibility provides another operational advantage. Authorized users can access Salesforce through a web browser from different locations, provided they have appropriate permissions and internet connectivity. This provides greater flexibility than a system restricted to a single local computer.

Security and access control are also relevant to operational use. Different users can be provided with appropriate permissions based on their responsibilities. This helps ensure that sensitive placement information is accessible only to authorized personnel.

Data quality must also be considered during operational use. Users should enter accurate information and update records regularly. Although validation rules can prevent certain types of incorrect data, proper administrative procedures are still required to maintain reliable placement information.

The system is also designed with future expansion in mind. If the institution later requires additional placement functionality, new modules can be incorporated. Potential extensions include interview scheduling, automated notifications, resume management, student self-service portals, advanced analytics, and AI-based job recommendations.

The system can therefore adapt to changing operational requirements without requiring a complete redesign of the entire application. This provides an advantage for institutions whose placement processes may evolve over time.

There are some operational dependencies. Users require appropriate Salesforce access and an internet connection to access the cloud-based application. The institution may also need to define user roles, data-entry procedures, and administrative responsibilities before deploying the system on a larger scale.

Despite these considerations, the Placement Tracker System is operationally feasible because it provides a centralized, accessible, structured, and user-friendly environment for placement management. It supports daily placement operations, reduces repetitive manual activities, improves access to information, and provides reporting and dashboard capabilities for monitoring and decision-making.

3.7 CHAPTER SUMMARY

This chapter presented a detailed requirement analysis of the proposed Placement Tracker System Using Salesforce CRM. The chapter established the requirements and resources necessary for developing, testing, deploying, and operating the proposed application.

The functional requirements of the system define the major operations that need to be supported, including student management, company management, application tracking, placement management, searching, reporting, and dashboard-based analysis. These requirements provide a clear understanding of the expected functionality of the application.

The non-functional requirements were also considered to ensure that the system provides appropriate security, usability, reliability, scalability, accessibility, and performance. These requirements are important for ensuring that the application remains useful and manageable as the amount of placement data increases.

The chapter presented the software requirements required for development. Salesforce CRM was identified as the primary application development platform, while Visual Studio Code and Salesforce CLI were used for coding, project management, metadata handling, and deployment. A modern web browser and stable internet connection were also identified as essential for accessing and testing the cloud-based application.

The hardware requirements of the system were also discussed. Since Salesforce provides cloud-based infrastructure, the proposed system does not require a dedicated local server or high-end computing infrastructure. A standard computer with adequate RAM, storage, processing capability, display, and internet connectivity is sufficient for development and testing.

The feasibility analysis evaluated the proposed system from technical, economic, and operational perspectives. The technical feasibility analysis demonstrated that Salesforce provides the required technologies for implementing the application, including Custom Objects, Object Relationships, Apex, SOQL, LWC, Validation Rules, Reports, and Dashboards.

The economic feasibility analysis considered the reduction in manual effort, paperwork, administrative workload, and infrastructure requirements. The centralized nature of the proposed system can reduce the time required for managing placement information and preparing reports.

The operational feasibility analysis demonstrated that the proposed system can support placement officers in their daily activities through a centralized and user-friendly interface. The LWC dashboard, search functionality, reports, and Salesforce Dashboard provide users with convenient access to placement information and analytical results.

Overall, the requirement and feasibility analysis indicates that the Placement Tracker System is technically, economically, and operationally feasible. The identified requirements and feasibility findings provide a strong foundation for the system design, technology selection, development, and implementation discussed in the subsequent chapters.

CHAPTER 4

TECHNOLOGY

4.1 INTRODUCTION

The **Placement Tracker System** has been developed using **Salesforce CRM**, a cloud-based platform that provides a secure, scalable, and flexible environment for developing enterprise applications. Salesforce provides several built-in services for data management, automation, security, application development, and reporting. Since the underlying infrastructure is managed by Salesforce, developers do not need to maintain dedicated servers or databases, allowing greater focus on implementing the functional and business requirements of the proposed system.

The project uses Salesforce Custom Objects to organize and store important placement-related information, including student, company, application, and placement records. Object Relationships are used to connect related records and maintain consistency among different entities. Validation Rules and Formula Fields are implemented to improve data accuracy and perform required calculations automatically.

For implementing business logic, the system uses Apex Programming and SOQL (Salesforce Object Query Language). Apex enables server-side processing and supports operations that require customized business logic, while SOQL is used to retrieve relevant records from Salesforce objects. Lightning Web Components (LWC) are used to develop the customized user interface and interactive placement dashboard.

The project also utilizes Reports and Dashboards for analyzing and presenting placement information through tables, charts, and key metrics. Salesforce CLI and Visual Studio Code are used as development and deployment tools for managing the Salesforce project, writing code, testing components, and deploying metadata.

Thus, the technologies used in the project collectively support the major aspects of the Placement

Tracker System, including data management, business logic, automation, user interface development, reporting, analytics, testing, and deployment. This chapter describes these technologies and explains their role in the implementation of the proposed system.

4.2 SALESFORCE PLATFORM

Salesforce is a cloud-based Customer Relationship Management (CRM) platform that enables organizations to develop, deploy, and manage enterprise applications through a centralized environment. It provides a secure and scalable platform where applications can be developed using both declarative tools and programming technologies such as Apex and Lightning Web Components (LWC).

A major advantage of Salesforce is its cloud-based architecture, which allows users to access applications and organizational data through an internet connection without requiring the installation of the complete application infrastructure on local systems. The platform also provides built-in capabilities for security, data management, backup, system availability, and platform updates, reducing the infrastructure management responsibilities of developers and organizations.

Salesforce supports both low-code/no-code configuration and programmatic development, making it suitable for applications with different levels of complexity. Administrators can configure objects, fields, relationships, validation rules, formulas, reports, and dashboards using declarative features, while developers can implement customized business logic and user interfaces using Apex, SOQL, and LWC.

For the Placement Tracker System, Salesforce serves as the primary development and deployment platform. It provides the environment required to manage student information, company details, job applications, and placement records in a centralized system. The platform also supports the implementation of relationships between these records, automation of placement-related processes, and generation of reports and dashboards.

The use of Salesforce therefore allows the proposed system to combine data management, application development, automation, reporting, and analytics within a single cloud-based environment. This makes Salesforce an appropriate platform for implementing the Placement Tracker System.

4.3 SALESFORCE ARCHITECTURE

Salesforce follows a multi-tenant architecture, in which multiple organizations use a common cloud infrastructure while maintaining logical isolation of their respective data and applications. Each organization operates within its own secure Salesforce environment, helping protect organizational information and maintain the confidentiality and integrity of stored data.

The architecture of Salesforce can be broadly understood through three major layers: the presentation layer, application layer, and database layer. These layers work together to provide the functionality required by enterprise applications such as the proposed Placement Tracker System.

The presentation layer is responsible for providing the user interface through which users interact with the application. Salesforce provides Lightning Experience and Lightning Web Components (LWC) for developing and displaying user interfaces. In the Placement Tracker System, the LWC dashboard provides users with an interactive interface for viewing student and placement information.

The application layer contains the business logic and rules that control how the application operates. Salesforce provides technologies such as Apex, Validation Rules, and Flows for implementing business processes and automation. In the proposed system, these technologies are used to validate data, process placement-related operations, and automate selected activities.

The database layer is responsible for storing and managing application data through Salesforce standard and custom objects. The Placement Tracker System uses custom objects to store information related to students, companies, applications, and placements. Relationships between objects allow related records to be connected and accessed efficiently.

The interaction between these layers enables the Placement Tracker System to provide centralized data management, business process automation, user-friendly interfaces, and reporting capabilities. Since Salesforce provides the underlying cloud infrastructure, the proposed system does not require separate server or database infrastructure, making the architecture suitable for developing a secure and scalable placement management application.

4.4 CUSTOM OBJECTS

Custom Objects are user-defined database structures created within Salesforce to store information specific to the requirements of an organization or application. Unlike standard Salesforce objects, custom objects can be designed with customized fields, relationships, validation rules, and other configurations according to the requirements of the proposed system.

In the Placement Tracker System, custom objects are used to create a structured data model for managing different stages of the placement process. The major custom objects implemented in the system are:

- Student
- Company
- Application
- Placement

The Student object stores information related to students, such as student identification details, contact information, branch, CGPA, eligibility, and placement status. The Company object maintains information about recruiting organizations, including company identification, package, location, eligibility criteria, required skills, and recruiter details.

The Application object is used to maintain information about students applying for company opportunities. It can connect a student with a company and store relevant application information such as application date, job role, and application status. The Placement object maintains information about students who successfully obtain placements, including the associated company, package, joining date, and related student information.

These custom objects are connected through appropriate object relationships, allowing information from different stages of the placement process to be accessed in a structured manner. Custom fields within each object store the specific information required by the application, while Salesforce features such as validation rules, formulas, Apex, and automation can be applied to these records.

Thus, the custom objects form the core data model of the Placement Tracker System and provide a structured foundation for storing, managing, tracking, and reporting placement-related information efficiently.

4.5 OBJECT RELATIONSHIPS

Object relationships establish logical connections between different Salesforce objects and allow related records to be associated with each other. They are an important part of the data model because they help maintain organized data and make it easier to retrieve related information.

The Placement Tracker System uses Lookup Relationships to connect the major placement-related objects. The relationships implemented in the system are:

- Student → Application
- Company → Application
- Student → Placement
- Company → Placement

The Student–Application relationship connects a student with their job applications, allowing the system to identify the companies and positions for which a particular student has applied. Similarly, the Company–Application relationship connects applications with the respective recruiting company.

The Student–Placement relationship connects a placement record with the student who has been placed. The Company–Placement relationship identifies the organization offering the placement. Together, these relationships allow the system to maintain a complete connection between students, companies, applications, and final placement outcomes.

These relationships also improve data retrieval and reporting. For example, Salesforce Reports can display student details together with application or placement information, while dashboards can use related records to generate meaningful placement statistics.

Thus, object relationships provide the structural foundation that connects different components of the

Placement Tracker System and support organized data management, efficient information retrieval, and comprehensive reporting.

4.6 APEX PROGRAMMING LANGUAGE

Apex is a strongly typed, object-oriented programming language developed by Salesforce for implementing custom business logic on the Salesforce platform. Its syntax is similar to Java, making it suitable for developers who are familiar with object-oriented programming concepts. Apex executes on Salesforce servers, allowing developers to process data securely while interacting directly with Salesforce objects and platform services. It is specifically designed to work with the Salesforce environment and provides a programming layer for implementing functionalities that cannot be achieved through declarative configuration alone.

Apex plays an important role in Salesforce application development because it allows developers to customize the behavior of an application according to specific business requirements. While Salesforce provides many built-in features for managing data and configuring applications, complex requirements may require additional programming logic. Apex provides the necessary programming capabilities for handling such requirements and extending the functionality of the Salesforce platform.

One of the major advantages of Apex is its close integration with the Salesforce database. Developers can directly work with Salesforce objects and records through Apex code. This allows applications to retrieve, create, update, and process records according to predefined requirements. Apex can also interact with other Salesforce services, making it suitable for developing applications that require customized data processing and business operations.

Apex supports several important programming concepts, including classes, methods, variables, conditional statements, loops, collections, exception handling, interfaces, and inheritance. These features allow developers to organize application logic into reusable and maintainable components. Object-oriented programming concepts also make it possible to divide a large application into smaller logical units, where each class is responsible for a particular functionality.

In addition to ordinary classes, Apex provides support for different types of application processing.

Developers can create controllers for user interfaces, triggers for automatic record processing, batch classes for processing large amounts of data, scheduled classes for executing tasks at specified times, and other programmatic components according to application requirements. These capabilities make Apex suitable for both simple and complex Salesforce applications.

Another important characteristic of Apex is its execution within the Salesforce cloud environment. Unlike traditional programming languages where developers may need to configure a separate application server, Apex runs on Salesforce's infrastructure. This allows developers to focus on application functionality while the underlying Salesforce environment manages the execution infrastructure.

Apex also follows Salesforce's governor limits. Since Salesforce operates as a multi-tenant cloud platform, resources are shared between different organizations. Governor limits help ensure that a single application does not consume an excessive amount of shared resources. Therefore, developers need to write efficient Apex code and avoid unnecessary database operations.

The use of governor limits encourages developers to follow efficient programming practices. For example, database queries should be written carefully, unnecessary queries should be avoided, and records should be processed efficiently. This is particularly important when an application handles a large number of student, company, application, or placement records.

In the Placement Tracker System, Apex is used to implement the business logic required for managing placement-related information. The system contains different Salesforce objects for Student, Company, Application, and Placement records. Apex classes can interact with these objects and perform operations according to the requirements of the application.

Apex controller classes are particularly important for connecting the backend of the application with the Lightning Web Component interface. The LWC dashboard requires placement-related information such as the total number of students, companies, applications, and placements. Apex can retrieve this information from Salesforce and provide it to the LWC for presentation.

The Lightning Web Component is responsible mainly for displaying information and handling user interaction, while Apex is responsible for processing requests and retrieving the required records.

For example, when a user opens the Placement Tracker Dashboard, the LWC can request student information from an Apex controller. The Apex method can then execute an appropriate SOQL query to retrieve the required records from the Student object. The retrieved data is returned to the LWC, where it can be displayed in the dashboard.

Similarly, Apex can be used to retrieve company, application, and placement information. This allows different sections of the dashboard to display updated information based on the records stored in Salesforce. The integration between Apex and LWC therefore provides the foundation for creating a dynamic and interactive application interface.

Apex can also support search and filtering functionality in the Placement Tracker System. When a user searches for a student or applies a filter such as branch or placement status, the application can use Apex and SOQL to retrieve the relevant records. This avoids the need to manually search through all available records.

Another important role of Apex is processing business rules. A placement management application may require specific conditions to be applied when records are created or updated. Apex can implement such customized logic when declarative Salesforce features are not sufficient for the requirement.

Apex Triggers provide another mechanism for automatic processing. A trigger can execute when records are inserted, updated, deleted, or undeleted. In a placement management environment, triggers can be used to perform automatic operations related to placement records or other business requirements. This allows certain operations to occur without requiring users to perform them manually.

Apex also supports exception handling. Errors may occur during data processing, record retrieval, or other operations. Exception handling allows developers to manage such situations in a controlled manner rather than allowing the application to fail unexpectedly. Appropriate error handling can improve application reliability and provide more meaningful feedback during development and testing.

The use of Apex classes also improves the modularity of the Placement Tracker System. Instead of

placing all business logic inside a single component, different functionalities can be organized into separate classes and methods. For example, one controller can be responsible for retrieving student information, while another can handle dashboard-related operations.

This modular structure makes the application easier to maintain. If a particular functionality needs to be modified, the developer can make changes to the relevant class without unnecessarily changing unrelated components. It also makes the code easier to understand and test.

Apex supports reusable methods, which can reduce duplicate code. When the same type of operation is required at multiple places in the application, a reusable method can be created instead of implementing the same logic repeatedly. This improves maintainability and reduces the possibility of inconsistencies.

In the proposed Placement Tracker System, Apex therefore acts as an important connection between Salesforce data and the application's user interface. The custom objects provide the data storage structure, SOQL retrieves the required records, Apex processes the information, and LWC presents the results to users.

Apex also contributes to the scalability of the application by allowing additional backend functionality to be added as the project grows. If new requirements are introduced, additional methods, classes, or processing logic can be implemented without redesigning the complete system.

For example, future versions of the Placement Tracker System may require functionality for interview scheduling, automated notifications, resume processing, eligibility checking, or advanced placement analysis. Apex can provide the programming layer required for implementing such customized functionality.

Another advantage of Apex is its integration with other Salesforce platform features. Apex can work alongside Validation Rules, Formula Fields, Reports, Dashboards, Flows, Custom Objects, and Lightning Web Components. This allows developers to combine declarative and programmatic approaches according to the complexity of the requirement.

The use of Apex also supports the development of secure applications. Since Apex executes within

the Salesforce platform, developers can work with Salesforce security mechanisms and access controls. Appropriate security considerations can be incorporated into Apex code and application configuration according to organizational requirements.

Apex development also requires careful consideration of performance. Efficient SOQL queries, appropriate data structures, and optimized processing are important when dealing with large amounts of data. This is particularly relevant for placement systems because the number of student and application records can increase over time.

The development and testing of Apex code can be performed using Salesforce development tools such as Visual Studio Code and Salesforce CLI. These tools allow developers to create classes, modify code, deploy components, and manage the Salesforce project in a structured manner.

Visual Studio Code provides a convenient environment for writing Apex classes and viewing the complete project structure. Salesforce Extensions provide syntax support and integration with Salesforce development workflows. Salesforce CLI can then be used to deploy the completed Apex components to the Salesforce organization.

Apex also helps separate the application's business logic from its user interface. This separation is important for software maintainability because changes to backend processing do not necessarily require major changes to the frontend. Similarly, changes to the dashboard layout can be implemented without rewriting the complete backend logic.

The use of Apex in the project also demonstrates the application of object-oriented programming principles in a cloud-based development environment. Classes and methods provide a structured approach to implementing functionality, while reusable components can reduce development effort.

From a project development perspective, learning Apex also provided an understanding of how traditional programming concepts can be applied within a CRM platform. Although Salesforce provides many declarative features, Apex becomes important when the application requires customized processing or integration between different components.

The Placement Tracker System therefore demonstrates the practical use of Apex in a real-world

application scenario. Instead of using Apex only for theoretical programming exercises, the project uses it to retrieve placement information, connect the backend with LWC, implement business logic, and support dashboard functionality.

Overall, Apex is one of the core technologies used in the Placement Tracker System. It provides the programming and business-logic layer required to connect Salesforce data with the customized application interface. Its integration with Salesforce Objects, SOQL, Lightning Web Components, and other platform features makes it possible to develop a structured, interactive, and maintainable placement management application.

Thus, Apex plays a significant role in transforming Salesforce from a basic data management platform into a customized application development environment. In the proposed Placement Tracker System, it contributes to efficient data processing, modular development, backend functionality, dashboard integration, and future extensibility of the application.

4.7 SALESFORCE OBJECT QUERY LANGUAGE (SOQL)

Salesforce Object Query Language (SOQL) is a query language used to retrieve records from Salesforce objects. It is similar in purpose to Structured Query Language (SQL), but it is specifically designed for accessing data stored within Salesforce standard and custom objects. SOQL provides developers with a structured method for requesting only the information required by an application instead of retrieving all available data. It is an important part of Salesforce development because most applications require some form of data retrieval for displaying records, generating reports, performing calculations, or implementing business functionality.

SOQL is closely integrated with the Salesforce data model. Salesforce stores information in objects, and each object contains fields and records. SOQL allows developers to specify the object from which data should be retrieved and the fields that should be returned. For example, a developer can retrieve the name, branch, and CGPA of students without retrieving unrelated fields. This makes the query more focused and allows the application to work with only the required information.

One of the major advantages of SOQL is its ability to select specific fields. In a large Salesforce object, many fields may be available, but an application may require only a small subset of them.

Selecting only the necessary fields makes the data retrieval process more organized and reduces unnecessary processing.

SOQL also supports filtering conditions through the WHERE clause. Filtering allows developers to retrieve records that satisfy particular requirements. For example, student records can be filtered according to branch, CGPA, or placement status. Similarly, application records can be filtered according to application status or company.

SOQL can also be used to retrieve students according to academic criteria. For example, if the application needs to identify students having a particular minimum CGPA, a suitable condition can be applied. This can be useful when placement activities involve eligibility requirements specified by recruiting companies.

SOQL supports comparison operators that can be used for filtering records. Conditions can be created using values, ranges, and other supported criteria. This makes SOQL useful for applications where information needs to be retrieved according to specific business conditions.

Another important feature of SOQL is sorting. Developers can organize retrieved records according to selected fields. For example, student records can be arranged according to CGPA, while application records can be ordered according to application date. Sorting can make the information easier to understand when it is displayed to users.

SOQL also provides mechanisms for limiting the number of records returned by a query. This can be useful when the application requires only a specific number of records or when displaying a limited result set. Limiting unnecessary records can also help improve the efficiency of application operations.

Another important capability of SOQL is its support for relationship queries. Salesforce objects can be connected using relationships, and SOQL can retrieve information associated with these relationships. This feature is particularly important for the Placement Tracker System because Student, Company, Application, and Placement objects are logically connected.

For example, an Application record can contain references to both a Student and a Company. This

allows application information to be connected with the student who submitted the application and the company offering the opportunity. Similarly, a Placement record can be associated with the relevant student and company.

SOQL relationship queries can therefore support the retrieval of related information from connected records. This reduces the need to maintain duplicate information and allows the application to work with the relationships established in the Salesforce data model.

SOQL is also closely connected with Apex. SOQL is generally used inside Apex classes when application logic needs to retrieve Salesforce records. Apex executes the query, receives the returned records, and then processes the information according to the application's requirements.

For example, the dashboard may need to display the total number of students. Apex can execute an appropriate SOQL operation to obtain the required information. The result can then be provided to the LWC, where it can be displayed in a KPI card.

Similarly, information required for student tables can be retrieved using SOQL. The query can select fields such as student name, branch, CGPA, skills, and placement status. Apex can then return these records to the LWC for display.

SOQL also supports the search functionality of the Placement Tracker System. When a user searches for a student, the application can retrieve records matching the search requirement. This provides a more efficient alternative to manually browsing through all available student records.

Filtering can also be applied through the dashboard. For example, users may want to view students from a particular branch or students with a specific placement status. The required conditions can be incorporated into the data retrieval logic so that only relevant records are returned.

SOQL therefore contributes directly to the usability of the application. Instead of presenting all records at once, the system can retrieve information according to the user's requirements. This makes the dashboard more focused and easier to use. The Application object contains information such as the student, company, job role, applied date, and application status. Queries can retrieve this information for displaying application details or supporting analytical features.

Another advantage of SOQL is its integration with Salesforce security and data access mechanisms. The actual access to Salesforce data depends on the permissions and security configuration of the organization. Therefore, developers need to consider the appropriate security requirements while designing data retrieval functionality.

SOQL differs from traditional SQL in several ways because it is designed specifically for the Salesforce object model. Although both languages are used for querying data, SOQL works with Salesforce objects and relationships rather than traditional relational database tables in the same manner as SQL.

SOQL is primarily used for retrieving data. It is not intended to perform all the database operations associated with traditional SQL. Salesforce provides other mechanisms and programming features for creating, updating, and deleting records. Apex can be used to perform these operations when required.

In the Placement Tracker System, the use of SOQL demonstrates how a cloud-based application can retrieve structured information from Salesforce objects and use that information within a customized interface. It provides the necessary connection between stored placement records and application functionality.

The use of SOQL also supports modular development. Apex classes can contain separate methods for retrieving different types of information. For example, one method may retrieve student records, another may retrieve company records, and another may retrieve placement statistics. These methods can then be called by the relevant Lightning Web Components.

This approach improves maintainability because changes to a particular data retrieval requirement can be made within the relevant method rather than modifying the complete application. It also makes the application logic easier to understand and test.

SOQL can also support future enhancements of the Placement Tracker System. If additional functionality is introduced, such as interview tracking, eligibility analysis, advanced search, or recruitment analytics, additional queries can be developed to retrieve the required information from new or existing Salesforce objects.

For example, a future version of the system could retrieve students who satisfy a company's eligibility requirements by combining information such as branch, CGPA, and skills. The retrieved information could then be displayed through an LWC component or used by another application process.

Another possible use is retrieving placement trends based on company, branch, or recruitment cycle. Such information could support advanced analytical features and provide management with better insights into placement performance.

The use of SOQL therefore contributes to both the current functionality and future scalability of the system. Its ability to work with Salesforce Objects, filters, relationships, and application logic makes it an important technology in the proposed application.

In the Placement Tracker System, SOQL is not used independently. It works together with Apex, Salesforce Objects, Object Relationships, and Lightning Web Components. Salesforce Objects provide the data storage structure, relationships connect related records, SOQL retrieves the required information, Apex processes the data, and LWC presents the result to the user.

SOQL also contributes to the real-time nature of the customized dashboard. When the LWC requests current information through Apex and SOQL, it can retrieve the latest available records from Salesforce. This allows users to work with updated placement information rather than relying on manually prepared static summaries.

For example, when a new placement record is added to Salesforce, the relevant data can be retrieved through the application's Apex and SOQL logic. The dashboard can then display updated placement statistics when the relevant data is refreshed or retrieved again.

Therefore, SOQL plays an essential role in making the Placement Tracker System data-driven and interactive. It allows the application to respond to user requirements and retrieve relevant information from the centralized Salesforce environment.

Overall, Salesforce Object Query Language is a fundamental technology used in the development of the Placement Tracker System. It provides structured access to Student, Company, Application, and Placement records and supports, search, filtering, and data-processing requirements.

The integration of SOQL with Apex and Lightning Web Components enables the proposed system to provide users with relevant and organized placement information. Properly designed queries also help ensure efficient use of Salesforce resources and support reliable application performance.

Thus, SOQL acts as an important link between the Salesforce data layer and the application's business logic. Its use in the Placement Tracker System demonstrates how structured data retrieval can be combined with Apex and LWC to develop a centralized, interactive, and efficient cloud-based placement management application.

4.8 LIGHTNING WEB COMPONENTS (LWC)

Lightning Web Components (LWC) is Salesforce's modern framework for developing interactive and responsive user interfaces using standard web technologies such as HTML, CSS, and JavaScript. LWC follows a component-based development approach in which an application interface is divided into smaller, independent, and reusable components. This approach makes application development more organized and allows individual components to be developed, tested, modified, and maintained separately.

A Lightning Web Component generally consists of multiple files, with each file serving a specific purpose. The HTML file defines the structure and elements of the user interface, while the JavaScript file contains the component's behavior and application logic. The CSS file is used to customize the visual appearance of the component, including layout, spacing, fonts, and other styling properties. The XML configuration file defines the component's metadata and specifies where the component can be used within Salesforce.

LWC uses modern web standards and provides improved performance compared with the older Aura component framework. Because LWC is based on standard web technologies, developers familiar with HTML, CSS, and JavaScript can understand and develop Salesforce user interfaces more easily. The framework also supports efficient rendering and component reusability, which helps reduce unnecessary development effort.

One of the important features of LWC is its ability to communicate with Salesforce backend services. Lightning Web Components can interact with Apex classes to retrieve, process, and update

Salesforce records. This allows developers to separate the user interface from server-side business logic. Apex can retrieve data using SOQL, while the LWC can present that information to users in an organized and interactive manner.

In the Placement Tracker System, LWC has been used to develop the Placement Tracker Dashboard. The dashboard provides a customized interface for displaying important placement information. It includes summary cards for displaying statistics such as total students, companies, applications, and placements. Student information can also be presented through a structured table for easier viewing.

The LWC dashboard also provides search functionality that allows users to locate student information more efficiently. The component can communicate with Apex controllers to obtain live data from Salesforce and display the latest available information. This provides a more interactive experience compared with a static web page.

Another important advantage of LWC is its ability to provide a responsive user interface. The dashboard can be designed to adjust its layout according to different screen sizes, making it easier for users to access placement information through different devices and screen resolutions.

In the proposed system, LWC therefore acts as an important connection between the Salesforce data layer and the end-user interface. It provides a modern and user-friendly environment for viewing placement information while allowing the application to use Salesforce backend functionality.

Overall, Lightning Web Components contribute significantly to the usability and functionality of the Placement Tracker System. By combining HTML, CSS, JavaScript, Apex integration, and Salesforce data, LWC enables the development of a customized placement dashboard that provides users with quick and organized access to important placement information.

4.9 VALIDATION RULES

Validation Rules are declarative tools provided by Salesforce to ensure that users enter valid and consistent information into the system. A validation rule evaluates a specified condition whenever a record is created or updated. If the condition defined in the rule is satisfied, Salesforce prevents the record from being saved and displays an appropriate error message to the user.

Validation Rules are an important part of data quality management because they prevent invalid information from entering the Salesforce database. Without validation, users may accidentally enter incomplete, incorrect, or logically inconsistent information. Such incorrect records can later affect reports, dashboards, calculations, and other application processes.

One of the major advantages of Validation Rules is that they can be implemented without writing extensive program code. Salesforce provides a formula-based mechanism through which administrators and developers can define conditions according to application requirements. This makes validation a useful declarative feature for enforcing business requirements.

Validation Rules can be applied to different types of fields and conditions. For example, a system can verify whether a required value has been entered, whether a numerical value falls within an acceptable range, or whether two related fields satisfy a specific condition. Date-related conditions can also be validated to ensure that users do not enter logically incorrect dates.

In the Placement Tracker System, Validation Rules are used to improve the quality and consistency of placement-related information. Student records may require valid academic or contact information, while company records may require appropriate recruitment-related details. Application and placement records can also be checked against predefined conditions before they are saved.

For example, validation can help ensure that important information is not entered in an invalid format or that values satisfy the requirements defined for the placement process. This reduces the possibility of incorrect records being stored and subsequently used in reports or dashboards.

Validation Rules also reduce the need for manual verification. Instead of placement administrators having to check every record manually, Salesforce can automatically evaluate the defined conditions during record creation or modification. This improves efficiency and helps users identify errors at the point where they occur.

Another benefit is improved consistency across the system. When a rule is applied to an object, the same condition can be enforced for different users who create or modify records. This helps maintain a common data standard throughout the application.

In addition, Validation Rules support the reliability of reports and analytical components. Since reports and dashboards depend on the accuracy of stored data, preventing invalid information at the point of entry improves the reliability of the information presented to users.

In the Placement Tracker System, Validation Rules therefore form an important part of the data management strategy. They help ensure that student, company, application, and placement records satisfy predefined requirements before being stored.

Overall, Validation Rules improve the quality, consistency, and reliability of the Placement Tracker System. They provide an efficient declarative mechanism for enforcing business requirements and reducing data entry errors without requiring complex programming.

4.10 FORMULA FIELDS

Formula Fields are custom Salesforce fields that automatically calculate values based on formulas defined by the administrator or developer. Unlike ordinary fields in which users manually enter values, Formula Fields calculate their results dynamically using information available within the Salesforce record.

Formula Fields are useful when an application requires calculated, derived, or automatically generated information. The value of a Formula Field is recalculated whenever the underlying record information changes. This ensures that users can view the latest calculated value without manually updating the field.

Formula Fields can perform different types of operations depending on the requirements of the application. They can perform mathematical calculations, logical comparisons, text operations, and date-related calculations.

One of the main advantages of Formula Fields is that they reduce repetitive manual calculations. If a value can be calculated from existing fields, there is no need for users to enter and maintain the calculated value separately. This reduces data redundancy and decreases the possibility of human errors.

Formula Fields also improve data consistency. Since the value is generated automatically using a

predefined formula, different users will obtain the same result when the same underlying data is used. This is particularly useful in applications where calculated information is frequently required for reports and dashboards.

In the Placement Tracker System, Formula Fields can be used wherever derived information is required from existing placement-related data. For example, calculations or logical indicators related to student eligibility, academic information, or placement-related conditions can be generated using formulas instead of requiring users to calculate or enter them manually.

Formula Fields can also be useful for reporting purposes. A calculated value can be included in Salesforce Reports and used for analysis without requiring additional processing. This allows reports to present both original information and automatically calculated information in a structured format.

Another advantage is that Formula Fields are maintained by Salesforce automatically. When the source values used by the formula are changed, the Formula Field reflects the updated result. This makes the information dynamic and reduces the maintenance effort required from users.

Formula Fields can also support business logic in a declarative manner. For simple calculations and conditions, administrators can use formulas instead of creating additional Apex code. This can simplify the overall application architecture and reduce unnecessary programming.

In the Placement Tracker System, the use of Formula Fields complements other Salesforce features such as Validation Rules, Apex, Reports, and Dashboards. Together, these features help create a structured and efficient data management environment.

Therefore, Formula Fields play an important role in reducing manual calculations, maintaining consistent information, and providing dynamically calculated values. Their use in the Placement Tracker System helps improve data accuracy and provides users with up-to-date information derived from existing records.

4.11 REPORTS AND DASHBOARDS

Reports and Dashboards are important analytical features provided by Salesforce for organizing, summarizing, and visualizing application data. Reports present Salesforce records in a structured

format, while Dashboards provide graphical representations of information through charts, tables, and metrics. Together, they allow users to understand large amounts of data more easily.

Salesforce Reports provide several options for organizing information. Users can apply filters to display only relevant records, sort information according to specific fields, group records based on selected attributes, and summarize numerical information. These capabilities allow users to generate meaningful views of the data stored within Salesforce.

In the Placement Tracker System, multiple reports have been developed to support different areas of placement management. The Student Report provides information about students, including details such as student identification, branch, CGPA, eligibility, contact information, and placement status. The Company Report contains information about recruiting companies and relevant company details.

The Application Report provides information about student applications. It can display details such as the student, company, job role, application date, and application status. This helps placement personnel monitor the progress of applications and identify the current status of recruitment activities.

The Placement Report provides information about students who have received placements. It can include details such as student name, company, package, joining date, branch, CGPA, and other relevant placement information. This report can be used to monitor final placement outcomes.

Salesforce Dashboards use information from reports to provide graphical and summarized views of organizational data. In the Placement Tracker System, dashboard components have been created to display important placement information. These include students by branch, applications by status, placements by company, and total placements.

The Students by Branch component provides a visual representation of the distribution of students across different branches. The Applications by Status component helps users understand the number of applications at different stages of the recruitment process. The Placements by Company component provides an overview of how many students have been placed with different companies. The Total Placements metric provides a quick summary of the number of placement records available in the system. Such metrics are useful for management because they provide important information without requiring users to manually examine individual records.

Reports and Dashboards also support decision-making. Placement officers can use reports to identify students, applications, and placement records, while management can use dashboards to obtain a summarized view of placement activities. This can help institutions monitor recruitment performance and identify areas that may require attention.

In addition to the Salesforce Dashboard, the Placement Tracker System includes a customized Lightning Web Component dashboard. The LWC dashboard provides an application-oriented interface, while the Salesforce Dashboard focuses primarily on reporting and analytics.

Thus, Reports and Dashboards provide an important analytical layer for the Placement Tracker System. They transform stored Salesforce records into structured and visual information that can be used for monitoring, analysis, and decision-making.

4.12 SALESFORCE CLI (SFDX)

Salesforce Command Line Interface (CLI), commonly referred to as Salesforce CLI or SFDX in Salesforce development workflows, is a command-line tool used for developing, managing, testing, and deploying Salesforce applications. It provides developers with commands for interacting with Salesforce organizations and managing project source code from a local development environment.

One of the major advantages of Salesforce CLI is that it supports **source-driven development**. Salesforce metadata such as Apex classes, Lightning Web Components, objects, configuration files, and other project components can be maintained within a local Salesforce DX project. This provides a structured approach to application development and makes it easier to manage changes.

Salesforce CLI also supports authentication and authorization with Salesforce organizations. Developers can connect their local project to a Salesforce organization and perform different operations using CLI commands. This allows source files to be deployed from the local development environment to Salesforce.

Another important capability is metadata deployment and retrieval. Developers can deploy newly created or modified components to Salesforce and retrieve existing metadata from the organization.

During the development of the Placement Tracker System, Salesforce CLI was used to create and

manage the Salesforce project structure. The project was developed as a Salesforce DX project containing the required metadata and source files.

Salesforce CLI was also used for deployment activities. Apex classes, Lightning Web Components, metadata files, and other Salesforce components could be transferred from Visual Studio Code to the Salesforce organization through CLI commands. This reduced the need to manually create every development component through the Salesforce interface.

The CLI also supports testing and development workflows. Developers can execute relevant commands for validating and deploying Salesforce components and can use it as part of a structured development process.

Integration between Salesforce CLI and Visual Studio Code further improves productivity. Developers can edit source code in Visual Studio Code and then use the integrated terminal or Salesforce extensions to perform deployment and synchronization operations.

For the Placement Tracker System, Salesforce CLI therefore acted as an important development and deployment tool. It supported project creation, organization authentication, metadata management, deployment, and synchronization between the local project and Salesforce.

Overall, Salesforce CLI provides a systematic approach to Salesforce application development and makes source-based development more efficient. Its use in the Placement Tracker System helped manage the project components and deploy the developed application to the Salesforce environment.

4.13 VISUAL STUDIO CODE

Visual Studio Code (VS Code) is a lightweight and feature-rich source code editor developed by Microsoft. It supports multiple programming languages and provides an extensible development environment through extensions. For Salesforce development, VS Code can be enhanced using Salesforce Extensions, which provide features specifically designed for developing and managing Salesforce applications.

One of the main advantages of Visual Studio Code is its support for multiple development technologies within a single environment. Developers can work with Apex, HTML, CSS, JavaScript,

XML, and configuration files without requiring separate editors. This makes it suitable for Salesforce projects that contain different types of source files.

The Salesforce extensions for Visual Studio Code provide features such as syntax highlighting, code completion, code navigation, debugging support, and integration with Salesforce CLI. These capabilities make it easier for developers to write and maintain Salesforce source code.

In the Placement Tracker System, Visual Studio Code was used as the primary development environment. The Salesforce project was opened in VS Code, where different project components were organized according to the Salesforce DX project structure.

Apex classes were created and modified using Visual Studio Code. These classes contain the server-side business logic required by the application. VS Code was also used to develop Lightning Web Components, including the HTML, JavaScript, CSS, and XML configuration files required for the Placement Tracker Dashboard.

The integrated terminal in Visual Studio Code was particularly useful for executing Salesforce CLI commands. Developers could authenticate the Salesforce organization, deploy source code, retrieve metadata, and perform other Salesforce operations without leaving the development environment.

Visual Studio Code also supports error detection and code assistance. Salesforce extensions and language services can identify syntax issues and provide suggestions while developers write code. This helps reduce development errors and improves coding efficiency.

Another advantage is its support for project organization. The Placement Tracker System contains multiple Salesforce components, including Apex classes, Lightning Web Components, objects, and metadata files. VS Code provides a file explorer and project structure that makes these components easier to locate and manage.

The editor also supports source control integration, which can be useful when maintaining the project in GitHub. Developers can track file changes, create commits, and manage project versions through integrated source control features.

Overall, the combination of Visual Studio Code and Salesforce CLI provided an efficient

development workflow for the proposed system. VS Code served as the primary coding environment, while Salesforce CLI provided the required command-line capabilities for authentication, deployment, and project management.

4.14 CHAPTER SUMMARY

This chapter discussed the major technologies used in the development of the **Placement Tracker System using Salesforce CRM**. It explained the role of Lightning Web Components, Validation Rules, Formula Fields, Reports and Dashboards, Salesforce CLI, and Visual Studio Code in the implementation of the system.

Lightning Web Components were used to develop the customized placement dashboard, while Validation Rules helped maintain data accuracy and consistency. Formula Fields provided dynamically calculated information, reducing manual calculations and data redundancy. Reports and Dashboards supported placement analysis by presenting student, application, company, and placement information in structured and graphical formats.

Salesforce CLI and Visual Studio Code provided the development and deployment environment for managing Salesforce source code, Apex classes, LWC components, and metadata. Together, these technologies supported the development of a centralized, scalable, and efficient Placement Tracker System.

CHAPTER 5

CODING

5.1 INTRODUCTION

The coding phase is one of the most significant stages in the Software Development Life Cycle (SDLC), where the system design and requirements are transformed into a fully functional software application. During this phase, all the modules identified in the requirement analysis are implemented using appropriate programming languages, development tools, and technologies. The success of a software project largely depends on the quality of implementation, coding standards, and proper integration of different modules.

The Placement Tracker System was developed using the Salesforce CRM platform, which provides a secure and cloud-based environment for application development. Unlike traditional software development, Salesforce offers both declarative and programmatic development approaches. Declarative features such as Custom Objects, Validation Rules, Formula Fields, Reports, and Dashboards reduce the amount of coding required, while Apex Programming and Lightning Web Components provide flexibility for implementing custom business logic and interactive user interfaces.

The development process followed a modular approach in which individual modules such as Student Management, Company Management, Application Management, and Placement Management were developed independently and later integrated into a single application. This approach simplified debugging, testing, and maintenance while improving the overall scalability of the system.

Visual Studio Code was used as the primary development environment, while Salesforce CLI was used for project creation, deployment, and synchronization with the Salesforce organization. Apex classes were developed to implement business logic, SOQL queries were used for data retrieval, and Lightning Web Components were created to build a responsive user interface. Salesforce Reports and Dashboards were also developed to provide analytical insights into placement activities.

5.2 PROJECT STRUCTURE

The Placement Tracker System follows the **Salesforce DX (SFDX) project structure**, which provides an organized framework for developing, maintaining, testing, and deploying Salesforce applications. The structure separates different types of Salesforce components into dedicated directories, making the project easier to understand, manage, modify, and extend. This standardized structure also helps developers maintain consistency throughout the development lifecycle.

The project was created using Salesforce CLI within Visual Studio Code. The Salesforce DX structure allows the source code and metadata of the application to be maintained in a standardized format. This approach supports source-driven development and makes it easier to synchronize the local project with the Salesforce organization. Developers can make changes locally and deploy the required components to the Salesforce environment when required.

The main project components are organized under the `force-app/main/default` directory. Different Salesforce components are maintained in their respective folders according to their type. Apex classes are stored in the `classes` directory, while Lightning Web Components are maintained inside the `lwc` directory. Salesforce custom objects and their related metadata are stored in the `objects` directory.

The `classes` directory contains the Apex classes used to implement the server-side business logic of the Placement Tracker System. These classes are responsible for operations such as retrieving student information, processing application and placement data, and providing data to Lightning Web Components. Each Apex class is generally accompanied by its corresponding metadata file, allowing Salesforce to recognize and deploy the class correctly.

The `lwc` directory contains the Lightning Web Components used to develop the customized user interface. Each LWC is maintained in a separate folder containing the required HTML, JavaScript, CSS, and XML configuration files. In the Placement Tracker System, the `studentDashboard` component is used to provide an interactive interface for displaying student and placement information.

The objects directory contains the metadata associated with the custom objects implemented in the system. The major custom objects include Student, Company, Application, and Placement. Each object can contain its own fields, relationships, validation rules, and other metadata required by the application. This organization provides a structured approach to maintaining the application's data model.

The project also contains configuration files that define the Salesforce project and its deployment settings. The `sfdx-project.json` file provides important information about the Salesforce DX project, including the project structure and source directories. The `package.xml` file can be used to specify the Salesforce metadata components that need to be retrieved from or deployed to a Salesforce organization.

The project structure also supports the separation of application logic, user interface components, and data configuration. Apex classes handle server-side processing, LWC components handle the presentation layer, and Salesforce objects manage the application's data. This separation makes the application easier to understand and reduces unnecessary dependency between different components.

Another important advantage of the SFDX structure is that it supports modular development. Individual components can be created, modified, tested, and deployed without requiring changes to the entire project. For example, modifications to the Student Dashboard can be performed within the LWC directory without directly affecting the custom object configuration.

The structure also simplifies the process of identifying and resolving development issues. If an error occurs in the server-side business logic, developers can examine the relevant Apex class. Similarly, user-interface problems can be investigated within the corresponding LWC files.

Salesforce DX also supports version control through tools such as Git and GitHub. Since the project components are maintained as source files, changes can be tracked and managed using a version control system. This is particularly useful for maintaining different versions of the Placement Tracker System and documenting changes made during development.

The project structure also supports future expansion of the system. Additional Apex classes, Lightning Web Components, custom objects, reports, and other Salesforce components can be added

to their appropriate directories without changing the overall organization of the project. Therefore, the structure provides flexibility for implementing future features such as interview management, notifications, advanced analytics, and additional placement modules.

Figure 5.1: Salesforce DX Project Struct

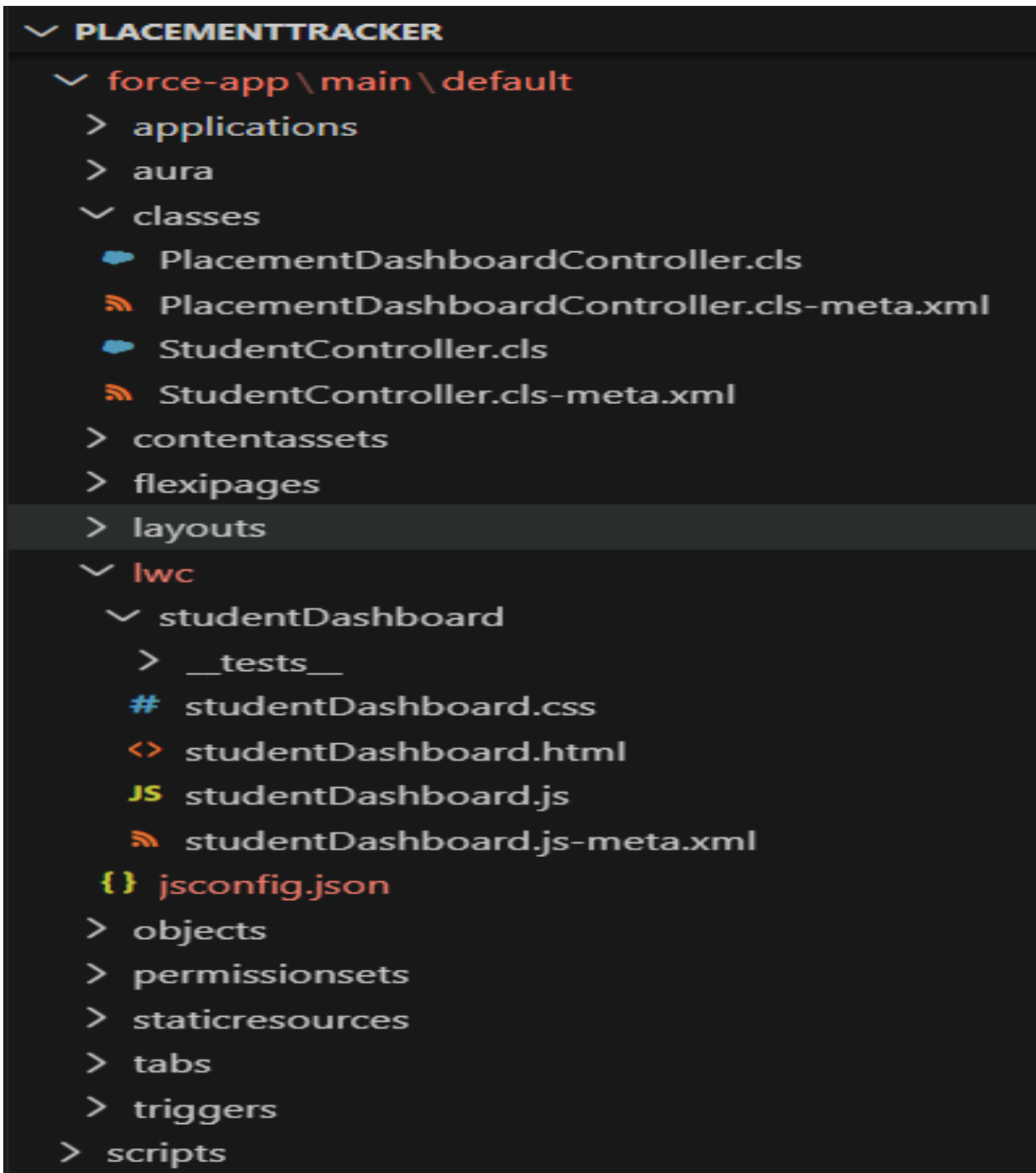


Table 5.1 Description of Project Folders

Folder	Description
classes	Stores Apex classes implementing business logic.
lwc	Contains Lightning Web Components developed for the user interface.
objects	Stores metadata of custom objects and custom fields.
reports	Contains Salesforce reports created for placement analysis.
dashboards	Stores dashboard components and analytics.
layouts	Defines page layouts for custom objects.
applications	Contains Lightning application metadata.
manifest	Stores deployment metadata files.

The modular project structure improves code organization and simplifies future maintenance. It also enables developers to deploy only the required components without affecting the remaining application.

5.3 DATABASE DESIGN (CUSTOM OBJECTS)

Database design is an important stage in application development because it determines how information will be stored, organized, related, and retrieved within the system. A well-designed database structure helps maintain data consistency, reduces unnecessary duplication, and supports efficient retrieval of information. Since Salesforce is a cloud-based platform, it does not expose traditional database tables directly to developers. Instead, Salesforce uses Standard Objects and Custom Objects to provide a structured data management environment.

For the Placement Tracker System, Custom Objects were created according to the requirements of the application to store and manage placement-related information. The major custom objects used in the system are Student, Company, Application, and Placement. Each object represents a specific entity involved in the placement process and contains fields designed to store the required information.

The Student object is used to maintain information about students participating in the placement process. It contains details such as Student ID, Student Name, Branch, CGPA, contact information, eligibility, and placement status. These fields allow placement officers to maintain complete student profiles within the system.

The Company object stores information about organizations participating in the recruitment process. Important information includes Company ID, Company Name, Package, Location, Eligibility CGPA, Required Skills, HR details, contact information, and Last Apply Date. This object helps maintain a centralized record of recruiting companies and their requirements.

The Application object maintains information about student applications submitted for different company opportunities. It connects students with companies and stores information such as Application ID, Applied Date, Status, Job Role, and other relevant application details. This allows the progress of student applications to be tracked throughout the recruitment process.

The Placement object stores information about successful placement outcomes. It maintains details such as Placement ID, Student, Company, Package, Joining Date, and related student information. This object represents the final stage of the placement process and allows successful placement records to be maintained separately.

Relationships between these custom objects establish logical connections between different entities. For example, a student can have application records associated with companies, while successful placement records connect students with the companies where they are placed. These relationships allow related information to be retrieved and displayed together in Salesforce Reports, Dashboards, and Lightning Web Components.

The database design also supports data validation and consistency through Salesforce features such as Validation Rules, Formula Fields, and Object Relationships. These features help ensure that the information entered into the system follows the requirements defined for the Placement Tracker System.

Overall, the custom object-based database design provides a structured foundation for the application. It allows student, company, application, and placement information to be stored in a centralized environment while supporting relationships, data validation, reporting, and future expansion of the system.

5.3.1 Student Object

The Student object is used to store the academic and personal details of students participating in campus placements. It acts as the primary source of student-related information required throughout the placement process.

The object contains fields such as Student ID, Student Name, Email Address, Phone Number, Branch, CGPA, Skills, and Placement Status. These fields help placement officers verify student eligibility, maintain academic records, and monitor placement progress.

The Student object serves as the foundation for other modules because both the Application and Placement objects maintain relationships with student records.

The object also helps in identifying students who are eligible for specific placement opportunities based on their academic and skill-related information.

Student records can be accessed by authorized users through Salesforce interfaces, reports, and dashboards.

The stored information can also be used to support application tracking and placement analysis.

Validation Rules can be applied to ensure that important student information is entered correctly and consistently.

Thus, the Student object provides a centralized and organized source of student information for the complete placement management process.

Figure 5.2: Student Custom Object

FIELD LABEL	FIELD NAME	DATA TYPE
Branch	Branch__c	Picklist
CGPA	CGPA__c	Number(3, 2)
Created By	CreatedById	Lookup(User)
Eligible for Placement	Eligible_for_Placement__c	Formula (Text)
Email	Email__c	Email
Last Modified By	LastModifiedById	Lookup(User)
Owner	OwnerId	Lookup(User,Group)
Phone	Phone__c	Phone
Placement Status	Placement_Status__c	Picklist
Skills	Skills__c	Long Text Area(32768)
Student Name	Name	Text(80)

5.3.2 Company Object

The Company object stores information about organizations participating in campus recruitment. It enables placement officers to manage recruitment drives and maintain company-specific information in a centralized location.

The object contains fields such as Company Name, Package Offered, Job Location, Eligibility CGPA, Required Skills, HR Name, HR Email, and Last Application Date. Maintaining this

information digitally simplifies company management and improves coordination during recruitment activities.

The Company object helps placement officers track important recruitment requirements and identify suitable opportunities for eligible students.

The Last Application Date field helps monitor application deadlines and ensures that important recruitment dates are not missed.

Company records can be linked with Application and Placement records to maintain complete recruitment information.

The information stored in this object can also be used to generate company reports and placement analytics.

Thus, the Company object provides a centralized source for managing recruiting organizations and their placement-related requirements.

Figure 5.3: Company Custom Object

FIELD LABEL	FIELD NAME	DATA TYPE
Company Name	Name	Text(80)
Created By	CreatedById	Lookup(User)
Eligibility CGPA	Eligibility_CGPA__c	Number(3, 2)
Email	Email__c	Email
HR Name	HR_Name__c	Text(80)
Last Apply Date	Last_Apply_Date__c	Date
Last Modified By	LastModifiedById	Lookup(User)
Location	Location__c	Text(100)
Owner	OwnerId	Lookup(User,Group)
Package	Package__c	Currency(16, 2)
Phone	Phone__c	Phone
Required Skills	Required_Skills__c	Long Text Area(32768)

5.3.3 Application Object

The Application object records every job application submitted by students. It establishes the connection between students and companies by storing information about applications submitted for different recruitment drives.

This object contains fields such as Application ID, Student, Company, Job Role, Applied Date, and Application Status. Placement officers use this object to monitor application progress and verify recruitment activities.

The Application object helps track the current status of each application throughout the recruitment process. It allows placement officers to identify whether an application is submitted, shortlisted, selected, or rejected based on the available status information.

The relationship between the Student and Company records allows each application to be associated with the appropriate student and recruiting organization. The application information can also be used to generate reports and dashboards for analyzing recruitment activities.

Thus, the Application object plays an important role in maintaining a structured record of student participation in different company recruitment processes.

Figure 5.4: Application Custom Object

FIELD LABEL	FIELD NAME	DATA TYPE
Application Name	Name	Auto Number
Applied Date	Applied_Date__c	Date
Company	Company__c	Lookup(Company)
Created By	CreatedById	Lookup(User)
Job Role	Job_Role__c	Text(80)
Last Modified By	LastModifiedById	Lookup(User)
Owner	OwnerId	Lookup(User,Group)
Status	Status__c	Picklist
Student	Student__c	Lookup(Student)

5.3.4 Placement Object

The Placement object maintains the final placement details of students who have successfully secured employment through campus recruitment. It represents the final stage of the placement process and provides complete placement records required for institutional reporting and statistical analysis.

The object stores information such as Placement ID, Student Name, Company Name, Offered Package, Joining Date, and Placement Status. These fields help placement officers maintain accurate records of students who have been successfully placed and the organizations offering employment opportunities.

The Placement object is related to the Student and Company objects, allowing the system to identify the student who received the placement and the corresponding recruiting organization. This relationship also makes it easier to retrieve related information when generating reports.

Thus, the Placement object provides a centralized record of successful placement outcomes and plays an important role in monitoring and analyzing the overall performance of the campus placement process.

Figure 5.5: Placement Custom Object

FIELD LABEL	FIELD NAME	DATA TYPE
Company	Company__c	Lookup(Company)
Created By	CreatedById	Lookup(User)
Joining Date	Joining_Date__c	Date
Last Modified By	LastModifiedById	Lookup(User)
Offer Status	Offer_Status__c	Picklist
Owner	OwnerId	Lookup(User,Group)
Package	Package__c	Currency(16, 2)
Placement Number	Name	Auto Number
Student	Student__c	Lookup(Student)

5.3.5 Advantages of Using Custom Objects

The use of custom objects provides several advantages for the Placement Tracker System:

- Data is stored in a structured and organized manner.
- Information can be retrieved quickly using Salesforce queries.
- Relationships between objects improve data consistency.
- Reports and dashboards can be generated directly from stored records.
- Custom fields allow the application to meet institutional requirements.
- Future enhancements can be implemented without redesigning the database.

5.4 OBJECT RELATIONSHIPS

Object relationships define the association between different custom objects within Salesforce and play an important role in organizing the data model of the Placement Tracker System. They allow related records to remain connected and enable users to retrieve relevant information from multiple objects without maintaining duplicate data. A well-defined relationship structure also improves data consistency and simplifies reporting and data analysis.

In the Placement Tracker System, Lookup Relationships have been used to establish connections between the major custom objects. Lookup relationships provide flexibility because the related records can maintain their own independent existence while still being associated with other records. This is suitable for the placement system because students, companies, applications, and placements represent separate entities with their own information.

The Student object is related to the Application object so that each application can be associated with the student who submitted it. This relationship allows placement officers to view a student's application details and monitor the student's participation in different recruitment drives.

Similarly, the Company object is related to the Application object. This relationship identifies the

company for which a particular application has been submitted. By connecting students, applications, and companies, the system can provide a complete view of the recruitment process.

The Student object is also related to the Placement object. This allows each placement record to be associated with the student who successfully obtained the placement. The relationship makes it possible to retrieve relevant student information while viewing placement records.

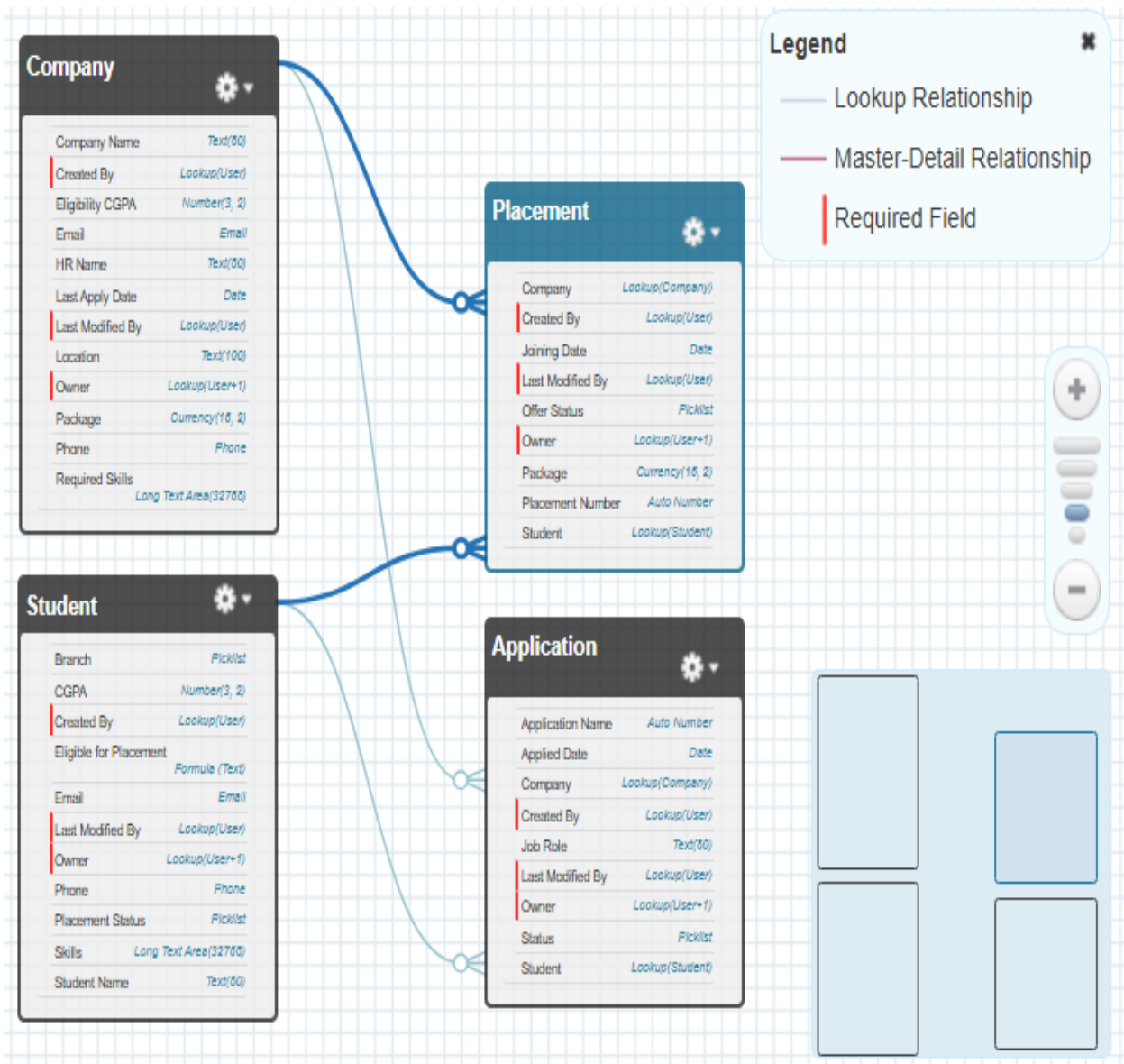
The Company object is also related to the Placement object. This connection identifies the organization that provided the placement opportunity. Together, the Student–Placement and Company–Placement relationships provide a complete representation of successful recruitment outcomes.

Table 5.2 Object Relationships

Parent Object	Child Object	Relationship
Student	Application	Lookup
Company	Application	Lookup
Student	Placement	Lookup
Company	Placement	Lookup

The use of object relationships provides several benefits. It reduces data redundancy, improves consistency, simplifies report generation, and enables users to retrieve related information efficiently. These relationships also support Salesforce Reports and Dashboards by allowing data from multiple objects to be displayed together.

Figure 5.6: Object Relationship Diagram



5.5 APEX CLASSES

Apex classes are the core components responsible for implementing the business logic of the Placement Tracker System. These classes process application requests, retrieve records from Salesforce objects, perform required operations, and provide the necessary information to Lightning Web Components. Apex executes on the Salesforce platform and allows the application to perform server-side processing while interacting directly with Salesforce data.

In this project, different Apex classes were developed to handle specific functionalities such as student management, company management, placement management, and dashboard operations. Dividing functionality into separate classes makes the application more organized and follows a modular development approach.

Apex classes interact with Salesforce custom objects using SOQL queries to retrieve the required records. For example, student-related classes can retrieve student information, while dashboard-related classes can retrieve data required for displaying placement statistics. The retrieved information can then be processed and returned to the Lightning Web Component for presentation.

The use of separate Apex classes also improves code reusability and maintainability. Each class is responsible for a specific area of functionality, making it easier to identify, modify, test, and maintain individual components without affecting the complete application.

The Apex classes also act as an important connection between the Salesforce database and the user interface. The Lightning Web Component can call appropriate Apex methods to obtain live Salesforce data and display it to users. This approach allows the dashboard to present updated information instead of relying only on static data.

Apex classes can also support exception handling and controlled processing of operations, helping the application respond appropriately when unexpected situations occur. In addition, Apex methods exposed to Lightning Web Components can be used to provide specific data required by different dashboard features.

The main Apex components developed for the Placement Tracker System include classes such as StudentService, StudentController, and PlacementDashboardController, along with other project-specific classes. These classes collectively support data retrieval, business logic, and communication between Salesforce records and the application's user interface.

Thus, Apex classes form an important part of the implementation of the Placement Tracker System. They provide a structured mechanism for implementing business logic, accessing Salesforce data, supporting LWC functionality, and maintaining a modular and maintainable application architecture.

Figure 5.7: Apex Classes

Open					
Entity Type	Entities		Related		
Entity Type	Name	Namespace ▲	Name	Extent	Direction
Classes	StudentService		← PlacementServiceTest	ApexClass	Referenced by
Triggers	PlacementService		Student	CustomFieldDefiniti...	References
Pages	CompanyService		Company	CustomFieldDefiniti...	References
Page Components	StudentServiceTest		Package	CustomFieldDefiniti...	References
Objects	CompanyServiceTest		Joining Date	CustomFieldDefiniti...	References
Static Resources	PlacementServiceTest		Offer Status	CustomFieldDefiniti...	References
Packages	PlacementTriggerTest		← Student__c	SObject	References
	ApplicationTriggerTest		← Student__c	SObject	References
	StudentController		← Company__c	SObject	References
	PlacementDashboardController		← Company__c	SObject	References
	DeveloperEditionUtils	devedapp	← Placement__c	SObject	References
	DeveloperEditionUtilsTest	devedapp	← Placement__c	SObject	References
	PostInstallScript	devedapp	← Placement__c	SObject	References
	PostInstallScriptTest	devedapp	← Placement__c	SObject	References

Open ☐ Filter ☐ Hide Managed Packages

5.5.1 StudentService

Purpose

The **StudentService** class is responsible for managing student-related operations within the Placement Tracker System. It retrieves student records, processes student information, and provides the required data to other modules of the application.

Working

The class communicates directly with the Student custom object. Whenever a user requests student information, the class retrieves the required records and returns them to the calling component. It also supports operations such as displaying student details and searching for student records.

Advantages

- Centralized management of student-related operations.
- Reduces duplicate code.
- Improves maintainability.
- Provides reusable methods for different modules.

5.5.2 CompanyService

Purpose

The **CompanyService** class manages company-related information within the application. It retrieves company records and provides recruitment-related information to users.

Working

The class interacts with the Company custom object to retrieve company details such as company name, package offered, eligibility criteria, and recruitment information. The retrieved data is displayed through reports and user interface components.

Advantages

- Simplifies company management.
- Improves data accessibility.
- Supports report generation.
- Enhances application performance.

5.5.3 PlacementService

Purpose

The **PlacementService** class manages placement information of selected students.

Working

The class retrieves placement records, processes placement statistics, and provides placement-related information to reports and dashboards. It also supports searching and displaying placement records.

Advantages

- Centralized placement management.
- Simplified record retrieval.
- Better report generation.
- Improved data consistency.

5.5.4 StudentController

Purpose

The **StudentController** class acts as an intermediary between the Lightning Web Components and Salesforce database.

Working

Whenever a Lightning Web Component requests student information, the StudentController invokes the required Apex methods and returns the retrieved data. It controls communication between the presentation layer and the database layer.

Advantages

- Separates presentation logic from business logic.
- Improves code organization.
- Supports reusable methods.
- Enhances application scalability.

5.5.5 PlacementDashboardController

Purpose

The **PlacementDashboardController** class retrieves statistical information required by the dashboard.

Working

The controller collects information such as the total number of students, companies, applications, and placements. It provides this information to Lightning Web Components for displaying KPI cards and dashboard statistics.

Advantages

- Provides real-time dashboard information.
- Improves dashboard performance.
- Supports graphical data visualization.
- Simplifies dashboard implementation.

5.5.6 Benefits of Using Apex Classes

The use of Apex classes offers several advantages during application development:

- Implements business logic efficiently.
- Improves code readability and maintainability.
- Supports modular application development.
- Enables seamless interaction with Salesforce data.
- Simplifies integration with Lightning Web Components.
- Facilitates future enhancements without affecting existing modules.

5.6 APEX TRIGGERS

Apex Triggers are special programs in Salesforce that execute automatically when records are inserted, updated, deleted, or undeleted. They are used to implement business logic that needs to run automatically whenever a specific database operation occurs. Unlike normal Apex methods, triggers are invoked by Salesforce based on predefined record events and do not require users to execute them manually.

In the Placement Tracker System, Apex Triggers are used to support automated processing of placement-related information and to ensure that predefined business rules are applied consistently. Triggers can work with the custom objects used in the system, such as Student, Company, Application, and Placement, depending on the requirements of the application.

Purpose of the Trigger

The trigger developed in this project is responsible for:

- Executing automatically during record operations.
- Validating placement-related information.
- Maintaining consistency among related records.
- Supporting automated processing of placement data.
- Reducing the need for repetitive manual operations.

The use of triggers allows the system to respond automatically whenever relevant records are modified. This is particularly useful in a placement management environment where changes to applications or placement records may require additional processing.

Working of the Trigger

Whenever a record is created, updated, deleted, or restored, Salesforce checks whether a trigger is associated with the respective object and event. If defined event occurs, Salesforce automatically

invokes the trigger. The trigger then evaluates the specified conditions and performs operations.

Triggers can execute either before or after a database operation. A before trigger can be used when information needs to be validated or modified before the record is saved. An after trigger can be used when an operation needs to be performed after the record has been successfully saved.

In the Placement Tracker System, this automated mechanism helps maintain consistency in placement-related records. For example, when relevant records are modified, the trigger can evaluate predefined conditions and perform the required processing without requiring manual intervention from the placement officer. The trigger therefore acts as an automated layer between record operations and business rules. It helps ensure that important processing is performed consistently whenever the associated Salesforce records are changed.

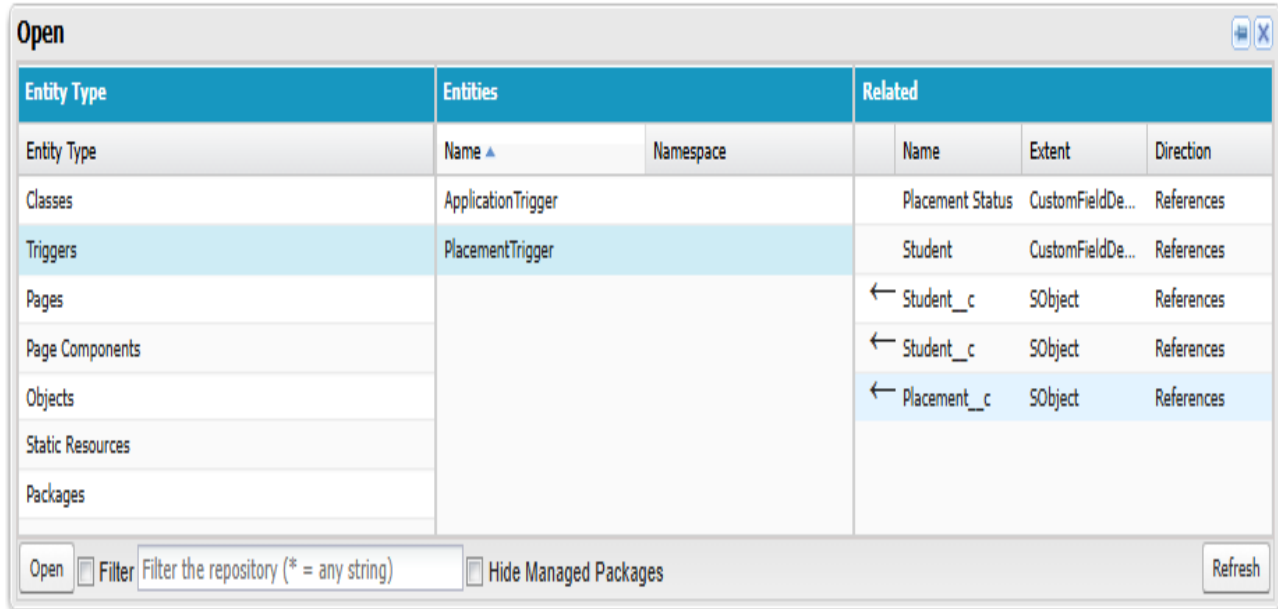
Advantages of Apex Triggers

The major advantages of using Apex Triggers in the Placement Tracker System include:

- Automates repetitive operations.
- Reduces manual effort.
- Improves application reliability.
- Enforces business rules automatically.
- Responds immediately to relevant record changes.
- Supports automated processing of related placement information.

Overall, Apex Triggers contribute to the automation and reliability of the Placement Tracker System. By executing automatically during relevant record operations, they help maintain consistent data and reduce dependency on manual processing. They complement other Salesforce features such as Validation Rules, Apex Classes, and Flows to provide a structured environment for implementing the business requirements of the application

Figure 5.8: Apex Trigger



Entity Type	Entities		Related		
Entity Type	Name ▲	Namespace	Name	Extent	Direction
Classes	ApplicationTrigger		Placement Status	CustomFieldDe...	References
Triggers	PlacementTrigger		Student	CustomFieldDe...	References
Pages			← Student_c	SObject	References
Page Components			← Student_c	SObject	References
Objects			← Placement_c	SObject	References
Static Resources					
Packages					

Open ☐ Filter Filter the repository (* = any string) ☐ Hide Managed Packages Refresh

5.7 LIGHTNING WEB COMPONENTS (LWC)

Lightning Web Components (LWC) is the modern user interface framework provided by Salesforce for developing responsive and high-performance web applications. It is based on standard web technologies such as HTML, CSS, and JavaScript, allowing developers to create reusable and interactive components. Compared to the older Aura framework, Lightning Web Components provide improved performance, enhanced security, and simplified development.

In the Placement Tracker System, Lightning Web Components were used to develop the main dashboard and user interface. The dashboard displays student records, placement statistics, search functionality, and other important information in an organized and user-friendly manner. The components communicate with Apex classes to retrieve real-time data from Salesforce objects and present it dynamically to the users.

Each Lightning Web Component consists of four files that work together to provide the complete functionality of the component.

5.7.1 HTML File

The HTML file defines the structure of the Lightning Web Component. It contains the layout of the dashboard, including cards, buttons, tables, input fields, and other interface elements. Salesforce Lightning Base Components were used to create a responsive and visually appealing user interface.

The HTML file organizes the information into different sections, allowing users to access placement-related information efficiently. The structured layout improves readability and enhances the overall user experience.

Figure 5.9: HTML File of Lightning Web Component

```
<!-- Dashboard Cards -->
<div class="dashboard">

  <div class="card students">
    <div class="icon"> 🧑 </div>
    <div>
      <h2>{totalStudents}</h2>
      <p>Total Students</p>
    </div>
  </div>

  <div class="card companies">
    <div class="icon"> 🏢 </div>
    <div>
      <h2>{totalCompanies}</h2>
      <p>Companies</p>
    </div>
  </div>

  <div class="card applications">
    <div class="icon"> 📄 </div>
    <div>
      <h2>{totalApplications}</h2>
      <p>Applications</p>
    </div>
  </div>

  <div class="card placements">
    <div class="icon"> 📍 </div>
    <div>
      <h2>{totalPlacements}</h2>
      <p>Placements</p>
    </div>
  </div>
</div>
```

5.7.2 JavaScript File

The JavaScript file controls the behavior of the Lightning Web Component. It handles user interactions, retrieves data from Apex classes, processes the received information, and updates the interface dynamically.

The JavaScript file establishes communication between the user interface and Salesforce server using Apex methods. It also manages events such as button clicks, search operations, and data refresh actions, ensuring that users always view updated information.

Figure 5.10: JavaScript File of Lightning Web Component

```
export default class StudentDashboard extends LightningElement {  
    columns = columns;  
    @track students = [];  
    @track filteredStudents = [];  
    totalStudents = 0;  
    totalCompanies = 0;  
    totalApplications = 0;  
    totalPlacements = 0;  
    selectedBranch = 'All';  
    selectedStatus = 'All';  
    wiredStudentsResult;  
    branchOptions = [  
        { label: 'All', value: 'All' },  
        { label: 'CSE', value: 'CSE' },  
        { label: 'IT', value: 'IT' },  
        { label: 'ECE', value: 'ECE' },  
        { label: 'ME', value: 'ME' }  
    ];  
    statusOptions = [  
        { label: 'All', value: 'All' },  
        { label: 'Placed', value: 'Placed' },  
        { label: 'Interview', value: 'Interview' },  
        { label: 'Applied', value: 'Applied' },  
        { label: 'Rejected', value: 'Rejected' }  
    ];  
};
```


5.7.3 CSS File

The CSS file is responsible for the visual appearance of the Lightning Web Component. It defines colors, fonts, spacing, alignment, borders, and responsive layouts that improve the presentation of the application.

Custom styling was implemented to enhance dashboard cards, tables, buttons, and search components. The use of CSS provides a professional appearance while ensuring that the dashboard remains responsive across different screen sizes.

Figure 5.11: CSS File of Lightning Web Component

```
.card p{
  color: #666;
  margin-top: 5px;
}

.students{
  border-left: 7px solid #0176d3;
}

.companies{
  border-left: 7px solid #2e844a;
}

.applications{
  border-left: 7px solid #fe9339;
}

.placements{
  border-left: 7px solid #8e44ad;
}

/* Search Card */
lightning-card{
  margin-bottom: 25px;
}

/* Company Summary */
.companySummary{
  display: flex;
  justify-content: space-around;
  padding: 25px;
}

.summaryBox{
  text-align: center;
}
```

5.7.4 Meta XML File

The Meta XML configuration file specifies how the Lightning Web Component is exposed within Salesforce. It defines the API version and the pages where the component can be used, such as Home Pages, App Pages, and Record Pages.

This configuration enables administrators to add the component using the Lightning App Builder without additional programming.

Figure 5.12: Meta XML Configuration File

```
<?xml version="1.0" encoding="UTF-8"?>
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
  <apiVersion>67.0</apiVersion>
  <isExposed>true</isExposed>

  <targets>
    <target>lightning__AppPage</target>
    <target>lightning__HomePage</target>
    <target>lightning__RecordPage</target>
  </targets>
</LightningComponentBundle>
```

5.7.5 Advantages of Lightning Web Components

The use of Lightning Web Components provides several benefits:

- Faster application performance.
- Component reusability.
- Modern and responsive user interface.
- Easy integration with Apex classes.
- Improved maintainability.
- Enhanced user experience.
- Better security through Salesforce framework.

5.8 SOQL QUERIES

Salesforce Object Query Language (SOQL) is the primary query language used for retrieving records from Salesforce objects. It enables efficient access to data stored in both standard and custom objects. Unlike traditional SQL, SOQL is specifically designed for the Salesforce platform and supports object relationships.

In the Placement Tracker System, SOQL is integrated within Apex classes to retrieve student details, company information, application records, and placement data. These records are displayed in Lightning Web Components, Reports, and Dashboards. SOQL also supports filtering and sorting of records, allowing users to access only the required information.

The efficient use of SOQL improves application performance by retrieving only the necessary data while maintaining the integrity and consistency of Salesforce records.

5.9 REPORTS

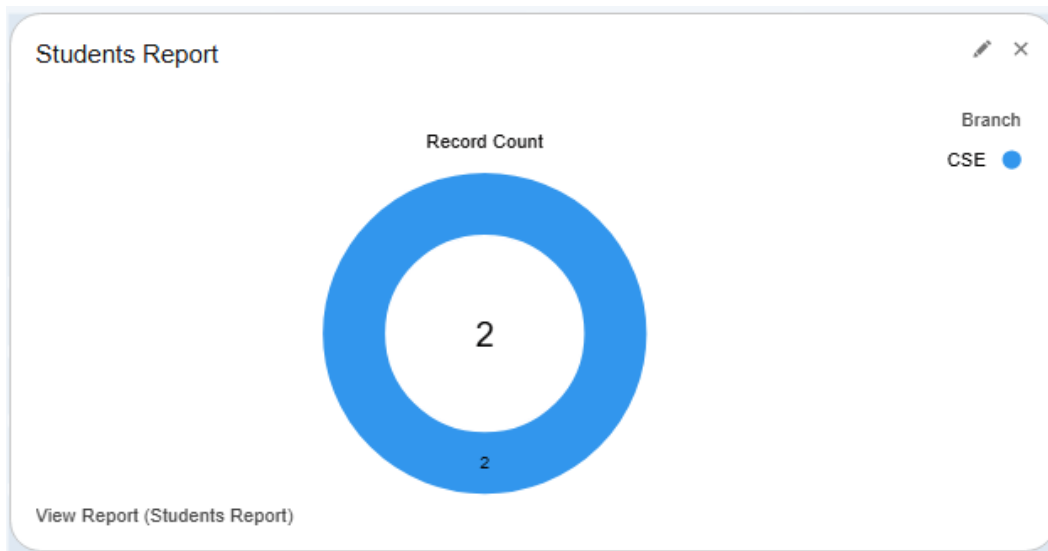
Reports are one of the most useful analytical features provided by Salesforce. They organize and summarize data stored within Salesforce objects, allowing users to analyze placement activities efficiently. Reports assist placement officers in monitoring student registrations, recruitment activities, applications, and final placements.

The Placement Tracker System includes three major reports.

5.9.1 Student Report

The Student Report provides complete information about registered students. It includes academic and personal details that help placement officers identify eligible candidates for recruitment drives. The report also supports filtering and sorting based on different criteria.

Figure 5.13: Student Report



5.9.2 Company Report

The Company Report displays details of recruiting organizations, including company name, package offered, eligibility criteria, location, HR contact information, and application deadlines. This report assists placement officers in managing recruitment activities efficiently.

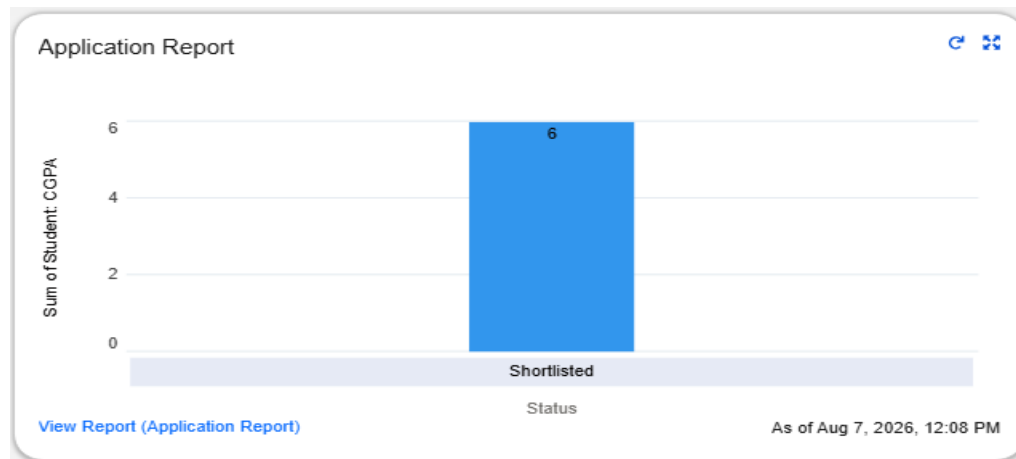
Figure 5.14: Company Report



5.9.3 Application Report

The Application Report contains information about applications submitted by students. It enables placement officers to monitor application progress and identify the current status of each application.

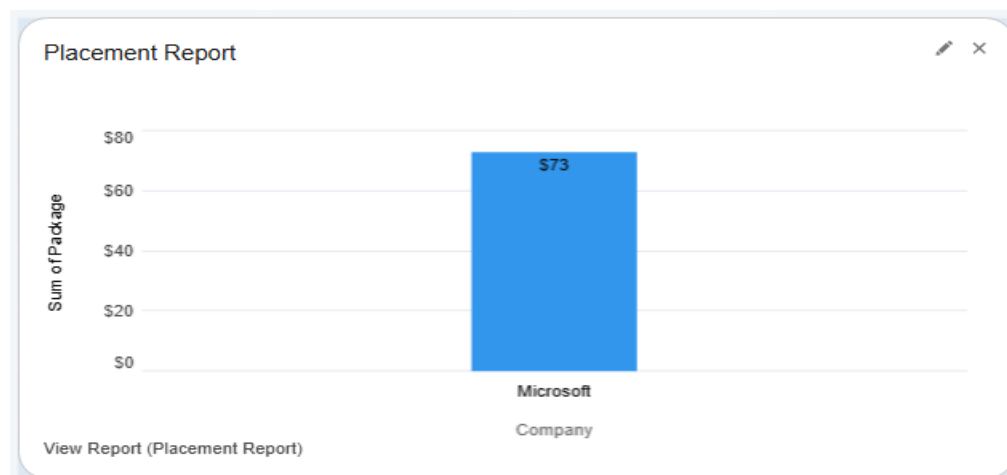
Figure 5.15: Application Report



5.9.4 Placement Report

The Placement Report displays details of successfully placed students, including company name, package, joining date, and placement status. It is useful for preparing institutional placement statistics and evaluating recruitment performance.

Figure 5.16: Placement Report



5.10 SALESFORCE DASHBOARD

The Salesforce Dashboard provides a graphical representation of placement-related information and helps users understand important placement statistics in a simple and organized manner. It is developed using Salesforce Reports and Dashboard components, which transform the stored placement data into visual charts, metrics, and summaries. The dashboard provides a centralized view of important information and reduces the need for users to examine individual records manually.

The dashboard created for the Placement Tracker System contains important Key Performance Indicators (KPIs) that provide a quick overview of the current placement situation. These include the total number of Students, Companies, Applications, and Placements. KPI components allow placement officers and management to view important figures immediately and understand the overall scale of placement activities.

In addition to KPI components, the dashboard contains graphical representations of placement-related data. The Students by Branch chart provides information about the distribution of students across different academic branches. This can help placement officers understand the number of students participating from different branches.

The Applications by Status chart provides an overview of application progress. It can be used to understand the number of applications at different stages of the recruitment process. This allows placement officers to monitor application activities and identify the current status of student applications.

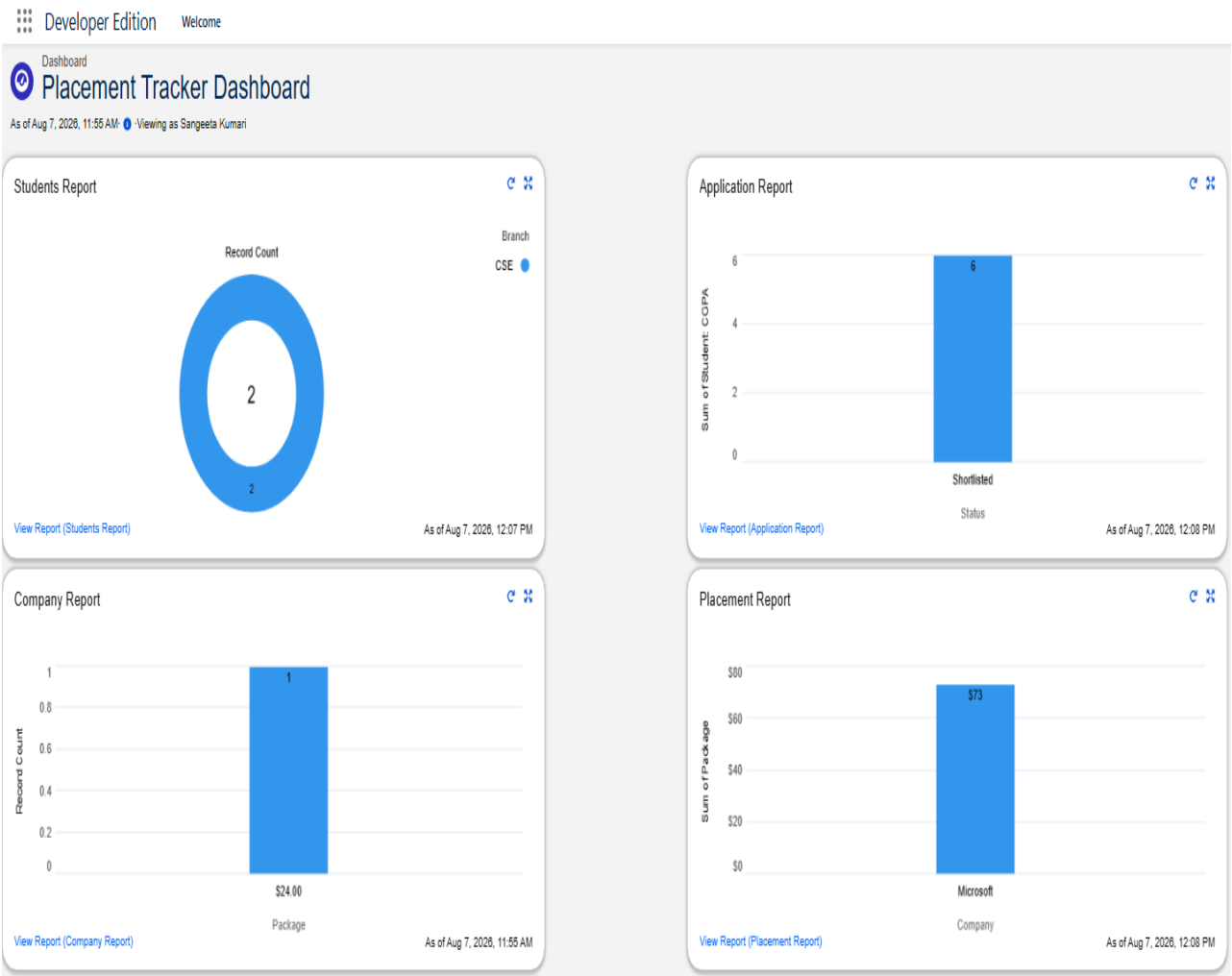
The Placements by Company chart provides a company-wise view of successful placements. It helps users identify which companies have recruited students and compare placement outcomes across different organizations. Such information can be useful for analyzing recruitment activities and preparing institutional placement statistics.

The dashboard is connected to Salesforce Reports, allowing its components to use the underlying placement data. When the report data is updated and the dashboard is refreshed, the dashboard components reflect the latest available information. This provides users with an updated view of placement statistics without requiring them to manually calculate the values.

The Salesforce Dashboard also complements the customized Lightning Web Component dashboard developed as part of the project. While the LWC dashboard provides a customized application interface for users, the Salesforce Dashboard focuses primarily on reporting, visualization, and analytical insights.

Overall, the Salesforce Dashboard provides an effective analytical interface for the Placement Tracker System. By combining KPI metrics, charts, and report-based information, it helps placement officers monitor recruitment activities, understand placement trends, and obtain important statistics efficiently. It therefore strengthens the reporting and decision-support capabilities of the proposed system.

Figure 5.17: Salesforce Dashboard



5.11 SYSTEM OUTPUT

The successful implementation of the Placement Tracker System is demonstrated through the final application interface developed using Lightning Web Components (LWC) on the Salesforce platform. The application integrates the major placement management functionalities into a centralized and interactive dashboard. This interface provides users with a convenient way to access and monitor information related to students, companies, applications, and placements.

The main dashboard displays Key Performance Indicator (KPI) cards representing the total number of students, companies, applications, and placements available in the system. These indicators provide an immediate overview of the current placement status and allow users to understand important statistics without manually reviewing individual records.

The application also provides a search facility that allows users to locate student records efficiently. Filtering options based on branch and placement status further improve the usability of the system by allowing users to narrow down the displayed records according to specific requirements. This functionality is particularly useful for placement officers when searching for eligible students or monitoring placement progress.

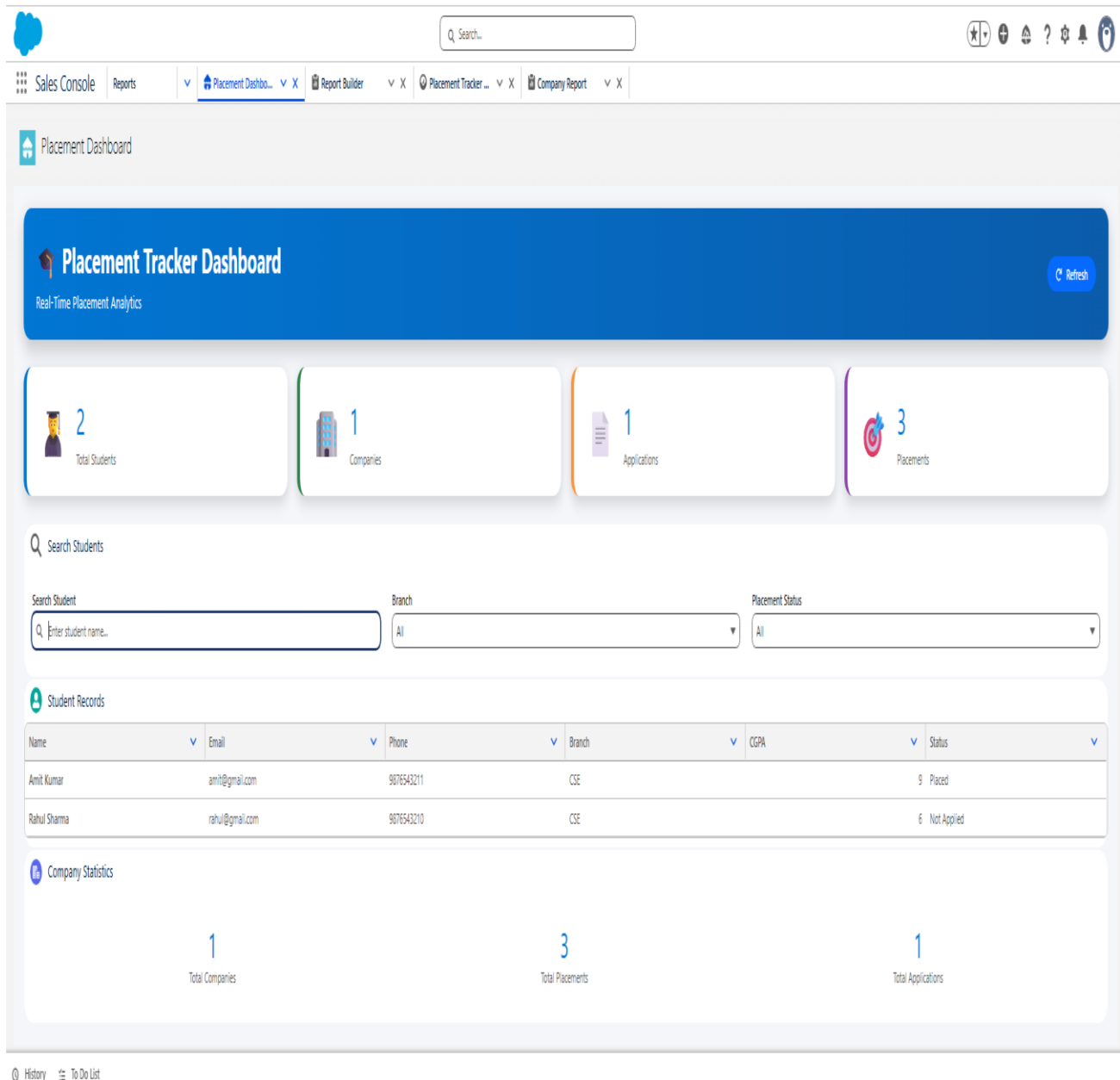
A dynamic Student Records table is included in the dashboard to display important academic and placement-related information. The table allows users to view relevant student details in an organized format. The dashboard also includes a Company Statistics section that provides a summarized view of company-related placement information.

The system retrieves data directly from Salesforce through Apex classes and SOQL queries. The Apex controller provides the required data to the Lightning Web Component, while JavaScript handles the presentation and interaction logic. This integration allows the dashboard to display updated Salesforce records and provides users with an interactive experience.

The final application successfully demonstrates the major functionalities implemented during the development process, including maintaining student and company records, managing applications and placements, retrieving Salesforce data, searching student information, applying filters, and displaying placement statistics.

The developed system therefore provides a centralized and user-friendly interface for placement management. It reduces dependence on manual record handling and enables placement officers to access important information efficiently. The final output demonstrates that the proposed Placement Tracker System successfully implements the planned functionality and provides a practical Salesforce-based solution for managing and monitoring campus placement activities.

Figure 5.18: Placement Tracker Dashboard (LWC)



5.12 CHAPTER SUMMARY

This chapter presented the implementation of the Placement Tracker System using Salesforce CRM. It discussed the Salesforce DX project structure and the design of custom objects used to manage students, companies, applications, and placements. The chapter also explained the relationships between these objects and the implementation of business logic using Apex classes and Apex Triggers.

The implementation of SOQL and Lightning Web Components was discussed, highlighting their role in retrieving Salesforce data and developing the interactive placement dashboard. The chapter also covered Salesforce Reports and Dashboards, which provide analytical information related to students, applications, companies, and placements.

Finally, the chapter presented the system output and demonstrated how the different Salesforce technologies work together to provide a centralized placement management solution. The successful integration of these components improves data organization, reduces manual effort, supports reporting and analysis, and provides an intuitive interface for placement officers. Overall, the implemented system provides a secure, scalable, and efficient environment for managing campus placement activities.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The Placement Tracker System developed using Salesforce CRM successfully achieves the objective of automating, organizing, and simplifying the campus placement management process. The system provides a centralized cloud-based platform for maintaining important placement-related information, including student details, company information, job applications, and placement records. By bringing these activities into a single environment, the system reduces the dependency on traditional manual methods and helps overcome issues related to scattered records, repetitive data entry, and difficulty in accessing updated placement information.

The implementation demonstrates the practical application of Salesforce CRM technologies for developing an enterprise-level placement management solution. Custom Objects were created to provide a structured data model for storing student, company, application, and placement information. Object Relationships were implemented to connect related records and allow information to be retrieved efficiently without unnecessary duplication. These components form the foundation of the system and support the complete placement lifecycle.

Apex Programming and SOQL were used to implement the server-side business logic and retrieve required information from Salesforce. Apex classes provide the processing layer between the Salesforce database and the user interface, while SOQL queries allow relevant records to be retrieved according to application requirements. Apex Triggers were also used to support automated processing and maintain consistency when relevant records are modified. The use of these technologies demonstrates how Salesforce can support both declarative configuration and programmatic development.

The project also makes extensive use of Lightning Web Components (LWC) to provide a customized and interactive user interface. The Placement Tracker Dashboard presents important information through KPI cards, student records, search functionality, and filtering options. The dashboard allows

placement officers to access important information in an organized manner and reduces the time required to locate relevant student or placement records. The responsive and structured interface improves usability and provides a more convenient way of interacting with the system.

Data quality and consistency were also considered during the development of the application. Validation Rules were implemented to ensure that records satisfy predefined conditions before being stored. Formula Fields were used where calculated information was required, reducing repetitive manual calculations and ensuring that derived values remain consistent with the underlying records. These Salesforce features help improve the reliability and accuracy of the information maintained by the system.

The implementation of Reports and Dashboards provides an additional analytical layer to the application. Student, Company, Application, and Placement Reports organize placement information into structured formats that can be filtered, grouped, and analyzed. The Salesforce Dashboard provides visual information such as students by branch, applications by status, placements by company, and total placement counts. These features allow placement officers and management to obtain useful insights from the stored data and support more informed decision-making.

The project also demonstrates the integration of development tools such as Salesforce CLI and Visual Studio Code. Salesforce CLI was used for project management, authentication, metadata operations, and deployment, while Visual Studio Code provided the development environment for Apex, LWC, configuration files, and other Salesforce components. The Salesforce DX project structure helped organize source code and metadata systematically and supported maintainability and version control.

From an operational perspective, the system reduces several repetitive activities associated with traditional placement management. Instead of maintaining separate spreadsheets or physical records for different placement activities, users can manage relevant information within a centralized Salesforce environment. The availability of search, filtering, reports, and dashboards further simplifies the process of monitoring recruitment activities.

The cloud-based nature of Salesforce also provides advantages in terms of accessibility and scalability. Since the application is hosted on the Salesforce platform, the project does not require

separate infrastructure for maintaining application servers and databases. This allows the system to be accessed through the Salesforce environment while relying on the platform's underlying infrastructure. The architecture also provides a foundation for expanding the application as the number of students, companies, applications, and placements increases.

Throughout the development process, software engineering principles and Salesforce development practices were applied to improve the organization and maintainability of the application. The separation of data management, business logic, and user interface components provides a modular structure. Custom Objects manage data, Apex handles business logic, SOQL retrieves information, and LWC provides the user interface. This modular approach makes individual components easier to understand, test, modify, and maintain.

The project also demonstrates the value of combining an operational dashboard with Salesforce analytical dashboards. The customized LWC dashboard provides an application-oriented interface for accessing placement information, while Salesforce Reports and Dashboards provide analytical views for monitoring placement performance. Together, these components provide both day-to-day operational support and higher-level reporting capabilities.

The developed system can therefore help educational institutions improve the organization of their placement activities. Placement officers can maintain student and company records, monitor applications, track placement outcomes, and generate reports from a centralized environment. The system also provides management with summarized placement statistics that can support monitoring and decision-making.

Although the current implementation fulfills the major requirements of the Placement Tracker System, the architecture provides opportunities for future enhancement. Additional functionality can be incorporated as institutional requirements evolve. Possible extensions may include more advanced analytics, interview management, automated notifications, additional recruitment workflows, enhanced student eligibility analysis, and integration with other institutional systems. The modular Salesforce architecture allows such features to be considered without requiring a complete redesign of the existing application.

The successful implementation of the project confirms that a Salesforce-based solution can provide a centralized, scalable, and user-friendly environment for managing placement activities. It improves data organization, reduces manual effort, supports analytical reporting, and provides placement officers with easier access to important information. Therefore, the Placement Tracker System serves as a practical example of how cloud-based CRM technology can contribute to the digital transformation and modernization of administrative processes within educational institutions.

6.2 FUTURE ENHANCEMENTS

Although the Placement Tracker System successfully fulfills the current project requirements, several additional features can be incorporated in future versions to further improve its functionality, automation, usability, and analytical capabilities. The proposed enhancements can help transform the system into a more comprehensive placement management platform capable of supporting students, placement officers, recruiters, and institutional management.

- **Integration with online job portals:** The system can be integrated with external job portals to automatically retrieve or publish job opportunities. This can reduce manual entry of company vacancies and make more opportunities available to students through a centralized platform.
- **Student portal:** A separate student portal can be developed through which students can log in, maintain their profiles, view eligible job opportunities, and directly submit applications. This would reduce the dependency on placement officers for routine application activities.
- **Email and SMS notifications:** Automated notification services can be implemented to inform students about application deadlines, interview schedules, shortlisting, selection results, and other placement-related updates.
- **Online aptitude and coding assessments:** The system can be integrated with online assessment platforms to conduct aptitude tests, coding assessments, and other recruitment evaluations. Assessment results could be associated with student records for more comprehensive recruitment analysis.

- **Interview scheduling module:** A dedicated interview management module can be introduced to allow placement officers and recruiters to schedule interviews, assign time slots, and monitor interview status. Students could receive automated notifications regarding their scheduled interviews.
- **Resume and document management:** Future versions can provide facilities for students to upload resumes, certificates, academic documents, and other recruitment-related files. Document verification and controlled access can also be incorporated to improve the recruitment workflow.
- **Artificial Intelligence-based recommendations:** AI techniques can be incorporated to recommend suitable job opportunities based on student skills, CGPA, branch, qualifications, and other relevant information. Such recommendations can help students identify opportunities that better match their profiles.
- **Alumni management:** An alumni module can be added to maintain information about previously placed students and establish communication with them. Alumni could potentially contribute through mentorship, career guidance, recruitment opportunities, and institutional engagement.
- **Mobile application:** A dedicated Android and iOS application can be developed to provide students and placement officers with convenient access to placement information through mobile devices. The application could provide notifications, job opportunities, application tracking, and placement updates.
- **Advanced analytics and predictive dashboards:** Future versions can include advanced analytical and predictive capabilities for identifying placement trends, analyzing company recruitment patterns, and estimating future placement outcomes.
- **Enhanced authentication and security:** Integration with external authentication services can provide additional security mechanisms and improve user authentication. Role-based access can also be extended to ensure that students, placement officers, recruiters, and administrators can access only the information relevant to their responsibilities.

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